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Fall protection compliance system and method

Abstract

A fall protection compliance system includes a connection structure configured for interfacing a connector comprising a frame and a connection sensor assembly. The connection structure includes a connection loop, a connection anchor, and a housing. The housing includes an identification element. The connection sensor assembly includes at least one short-range wireless communication antenna and a control device. The control device includes at least one processor which is configured to receive connection data from the short-range wireless communication antenna, determine a connection status of the connector based on the connection data and perform at least one action based on the connection status.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation in part of U.S. patent application Ser. No. 17/876,699, filed Jul. 29, 2022, which is a continuation in part of U.S. patent application Ser. No. 16/742,381, filed Jan. 14, 2020, which claims priority to U.S. Provisional Application No. 62/792,208, filed Jan. 14, 2019, U.S. Provisional Application No. 62/861,545, filed Jun. 14, 2019, and U.S. Provisional Application No. 62/944,071, filed Dec. 5, 2019, the entireties of each of which are incorporated by reference herein in their entirety.

BACKGROUND

Technical Field

(1) The present disclosure relates generally to the field of safety equipment, and more particularly, to systems and methods for monitoring and recording the use of fall protection equipment.

Technical Description

(2) Safety equipment is often required in the fields of construction, rescue operations, and recreational activities that occur in environments that pose a danger to the user, such as a risk of falling from an elevated height. Most safety equipment is subject to compliance with various regulatory standards, such as those instituted by the Occupational Safety and Health Administration (OSHA) and the National Institute for Occupational Safety and Health (NIOSH). These standards contain information regarding the use of safety equipment, as well as the need for proper inspection, monitoring, and record keeping.

(3) A number of challenges exist regarding inspection, monitoring, and record keeping for safety equipment. For example, verification of use and compliance of such use with existing standards is difficult to confirm without actual physical inspection of the individual wearing the safety equipment while performing the activity. Direct and ongoing verification of use for compliance is virtually impossible in environments where dozens of workers at a construction site may be using different types of safety equipment.

(4) In view of these and other disadvantages of existing safety equipment, it is desirable to provide systems and methods for monitoring and recording the use of safety equipment.

SUMMARY

(5) Therefore, and generally, the present disclosure provides improved systems and methods for monitoring and recording the use of safety equipment. In particular, the present disclosure provides

a fall protection compliance system and method that may be configured to detect that a connection has been made between a worker wearing a fall protection harness or similar safety equipment, and an anchor point. The system and method may be further configured to indicate the proper connection, disconnection, and usage compliance of the safety equipment regarding an unsafe or safe condition or area. The system and method may be further configured to store information regarding various aspects of the system, such as connection status of the safety equipment and/or length of use of the safety equipment. The system and method may be further configured to transmit connection status information to an external device or system.

(6) In some non-limiting embodiments or aspects, a connector for a fall protection compliance system may include a frame having an opening and a connection area configured for receiving a connection structure therein, and a connection sensor assembly associated with the frame. The connection sensor assembly may include at least one permanent magnet configured for generating a magnetic field within at least a portion of the connection area, at least one magnetometer configured for detecting a presence or an absence of a disturbance in the magnetic field, and a control device having at least one processor programmed or configured to: receive connection data from the at least one magnetometer, determine a connection status of the connector based on detecting the presence or the absence of the disturbance in the magnetic field, and perform at least one action based on the connection status.

(7) In some non-limiting embodiments or aspects, detecting the presence of the disturbance in the magnetic field may be indicative of a presence of a ferromagnetic material of the connection structure within the connection area, and detecting the absence of the disturbance in the magnetic field may be indicative of an absence of the ferromagnetic material of the connection structure within the connection area.

(8) In some non-limiting embodiments or aspects, the connection status may include a connected status indicative of a presence of a ferromagnetic material of the connection structure within the connection area and a disconnected status indicative of an absence of the ferromagnetic material of the connection structure within the connection area.

(9) In some non-limiting embodiments or aspects, the at least one action may include providing at least one of a visual indication, an audio indication, and a tactile indication based on the connection status. The at least one action may include providing at least one of a first visual indication, a first audio indication, and a first tactile indication indicative of a presence of a ferromagnetic material of the connection structure within the connection area, or providing at least one of a second visual indication, a second audio indication, and a second tactile indication indicative of an absence of the ferromagnetic material of the connection structure within the connection area. At least one of the first visual indication, the first audio indication, and the first tactile indication may be the same as or different from the second visual indication, the second audio indication, and the second tactile indication.

(10) In some non-limiting embodiments or aspects, the connection sensor assembly further may include a communication interface, and the at least one action may include transmitting the connection status to a remote device using the communication interface. The connection sensor assembly further may include at least one short-range wireless communication antenna configured for detecting a presence or an absence of an identification element on the connection structure, and at least one processor may be further programmed or configured to: receive communication data from the at least short-range wireless communication antenna, and determine the connection status of the connector based on detecting the presence or the absence of the identification element. Detecting the presence of the identification element may be indicative of a presence of an approved connection structure, and detecting the absence of the identification element may be indicative of an absence of the approved connection structure.

(11) In some non-limiting embodiments or aspects, the at least one processor may be further programmed or configured for storing the data received from the at least one magnetometer and the

at least short-range wireless communication antenna. The at least one magnet and the at least one magnetometer may be positioned away from a surface of the frame that is configured for contacting the connection structure.

(12) In some non-limiting embodiments or aspects, the connector may have a gate for selectively enclosing the opening of the frame. The gate may be movable between a closed position preventing passage through the opening and into the connection area and an open position permitting passage through the opening and into the connection area. The gate may be biased to the closed position by a biasing member. A locking assembly may be operatively connected to the gate and movable between a first position and a second position. In the first position, the locking assembly may be configured for preventing movement of the gate from the closed position toward the open position, and, in the second position, the locking assembly may be configured for permitting movement of the gate from the closed position toward the open position.

(13) In some non-limiting embodiments or aspects, a computer-implemented method for detecting connection of a connector to a connection structure may include receiving, with a control device having at least one processor, connection data gathered by at least one sensor configured for detecting a presence or an absence of an object within a connection area of the connector, determining, with the control device, a connection status of the connector based on detecting the presence or the absence of the object, and performing, with the control device, at least one action based on the connection status.

(14) In some non-limiting embodiments or aspects, detecting the presence of the object may be based on detecting a disturbance in a magnetic field within the connection area that is indicative of a presence of a ferromagnetic material of the connection structure within the connection area, and detecting the absence of the object may be based on detecting a disturbance in the magnetic field within the connection area that is indicative of an absence of the ferromagnetic material of the connection structure within the connection area.

(15) In some non-limiting embodiments or aspects, performing the at least one action may include providing at least one of a visual indication, an audio indication, and a tactile indication based on the connection status.

(16) In some non-limiting embodiments or aspects, the method may further include receiving, with the control device, communication data gathered by at least one short-range wireless communication antenna configured for detecting a presence or an absence of an identification element on the connection structure. Determining the connection status of the connector may be further based on detecting the presence or the absence of the identification element on the connection structure.

(17) In some non-limiting embodiments or aspects, a connector for a fall protection compliance system may include a frame having an opening and a connection area configured for receiving a connection structure therein, and a connection sensor assembly associated with the frame. The connection sensor assembly may have at least one permanent magnet configured for generating a magnetic field within at least a portion of the connection area, at least one magnetometer configured for detecting a presence or an absence of a disturbance in the magnetic field, at least one short-range wireless communication antenna configured for detecting a presence or an absence of an identification element on the connection structure, and a control device having at least one processor programmed or configured to: receive connection data from the at least one magnetometer and communication data from the at least short-range wireless communication antenna, determine a connection status of the connector based on detecting the presence or the absence of the disturbance in the magnetic field and based on detecting the presence or the absence of the identification element, and perform at least one action based on the connection status. Detecting the presence of the disturbance in the magnetic field may be indicative of a presence of a ferromagnetic material of the connection structure within the connection area. Detecting the absence of the disturbance in the magnetic field may be indicative of an absence of the

ferromagnetic material of the connection structure within the connection area. Detecting the presence of the identification element may be indicative of a presence of an approved connection structure, and detecting the absence of the identification element may be indicative of an absence of the approved connection structure.

(18) In some non-limiting embodiments or aspects, a connector for a fall protection compliance system may have a frame having an opening and a connection area configured for receiving a connection structure therein, and a connection sensor assembly associated with the frame. The connection sensor assembly may have at least one short-range wireless communication antenna configured for detecting a presence or an absence of an identification element on the connection structure, and a control device having at least one processor programmed or configured to: receive communication data from the at least short-range wireless communication antenna, determine a connection status of the connector based on detecting the presence or the absence of the identification element, and perform at least one action based on the connection status.

(19) In some non-limiting embodiments or aspects, detecting the presence of the identification element may be indicative of a presence of an approved connection structure, and detecting the absence of the identification element may be indicative of an absence of the approved connection structure. The connection status may include a connected status indicative of a presence of an approved connection structure, and a disconnected status indicative of an absence of the approved connection structure. The at least one action may include providing at least one of a visual indication, an audio indication, and a tactile indication based on the connection status. The connection sensor assembly further may include a communication interface, and wherein the at least one action includes transmitting the connection status to a remote device using the communication interface.

(20) As a further example, a fall protection compliance system includes a connection structure configured for interfacing an anchor structure and a connection sensor associated with the connection structure. The connection sensor includes an accelerometer or a pressure switch, where the connection sensor is configured to monitor continued connection of the connection structure to the anchor structure over a period of time and to output a signal indicating said continued connection.

(21) As another example, in a method for providing fall protection compliance monitoring, a signal from a connection sensor associated with a connection structure configured for interfacing with an anchor structure is received, where the connection sensor comprises an accelerometer or a pressure switch. A determination is made as to whether a user is continually connected over a period of time based on the signal, and a signal is output indicating said determination of whether the user is continually connected.

(22) Further non-limiting embodiments or aspects will now be described in the following numbered clauses.

(23) Clause 1. A connector for a fall protection compliance system, the connector comprising: a frame having an opening and a connection area configured for receiving a connection structure therein; and a connection sensor assembly associated with the frame, the connection sensor assembly comprising: at least one permanent magnet configured for generating a magnetic field within at least a portion of the connection area; at least one magnetometer configured for detecting a presence or an absence of a disturbance in the magnetic field; and a control device comprising at least one processor programmed or configured to: receive connection data from the at least one magnetometer; determine a connection status of the connector based on detecting the presence or the absence of the disturbance in the magnetic field; and perform at least one action based on the connection status.

(24) Clause 2. The connector according to clause 1, wherein detecting the presence of the disturbance in the magnetic field is indicative of a presence of a ferromagnetic material of the connection structure within the connection area, and wherein detecting the absence of the

disturbance in the magnetic field is indicative of an absence of the ferromagnetic material of the connection structure within the connection area.

(25) Clause 3. The connector according to clause 1 or 2, wherein the connection status comprises a connected status indicative of a presence of a ferromagnetic material of the connection structure within the connection area and a disconnected status indicative of an absence of the ferromagnetic material of the connection structure within the connection area.

(26) Clause 4. The connector according to any of clauses 1-3, wherein the at least one action includes providing at least one of a visual indication, an audio indication, and a tactile indication based on the connection status.

(27) Clause 5. The connector according to any of clauses 1-4, wherein the at least one action includes providing at least one of a first visual indication, a first audio indication, and a first tactile indication indicative of a presence of a ferromagnetic material of the connection structure within the connection area, or providing at least one of a second visual indication, a second audio indication, and a second tactile indication indicative of an absence of the ferromagnetic material of the connection structure within the connection area.

(28) Clause 6. The connector according to any of clauses 1-5, wherein at least one of the first visual indication, the first audio indication, and the first tactile indication is the same as the second visual indication, the second audio indication, and the second tactile indication.

(29) Clause 7. The connector according to any of clauses 1-6, wherein at least one of the first visual indication, the first audio indication, and the first tactile indication is different from the second visual indication, the second audio indication, and the second tactile indication.

(30) Clause 8. The connector according to any of clauses 1-7, wherein the connection sensor assembly further comprises a communication interface, and wherein the at least one action includes transmitting the connection status to a remote device using the communication interface.

(31) Clause 9. The connector according to any of clauses 1-8, wherein the connection sensor assembly further comprises at least one short-range wireless communication antenna configured for detecting a presence or an absence of an identification element on the connection structure, and wherein at least one processor is further programmed or configured to: receive communication data from the at least short-range wireless communication antenna; and determine the connection status of the connector based on detecting the presence or the absence of the identification element.

(32) Clause 10. The connector according to any of clauses 1-9, wherein detecting the presence of the identification element is indicative of a presence of an approved connection structure, and wherein detecting the absence of the identification element is indicative of an absence of the approved connection structure.

(33) Clause 11. The connector according to any of clauses 1-10, wherein the at least one processor is further programmed or configured for storing the data received from the at least one magnetometer and the at least one short-range wireless communication antenna.

(34) Clause 12. The connector according to any of clauses 1-11, wherein the at least one magnet and the at least one magnetometer are positioned away from a surface of the frame that is configured for contacting the connection structure.

(35) Clause 13. The connector according to any of clauses 1-12, further comprising a gate for selectively enclosing the opening of the frame, wherein the gate is movable between a closed position preventing passage through the opening and into the connection area and an open position permitting passage through the opening and into the connection area.

(36) Clause 14. The connector according to any of clauses 1-13, wherein the gate is biased to the closed position by a biasing member.

(37) Clause 15. The connector according to any of clauses 1-14, further comprising a locking assembly operatively connected to the gate and movable between a first position and a second position, wherein, in the first position, the locking assembly is configured for preventing movement of the gate from the closed position toward the open position, and wherein, in the second position,

the locking assembly is configured for permitting movement of the gate from the closed position toward the open position.

(38) Clause 16. A computer-implemented method for detecting connection of a connector to a connection structure, the method comprising: receiving, with a control device comprising at least one processor, connection data gathered by at least one sensor configured for detecting a presence or an absence of an object within a connection area of the connector and; determining, with the control device, a connection status of the connector based on detecting the presence or the absence of the object, and performing, with the control device, at least one action based on the connection status.

(39) Clause 17. The method according to clause 16, wherein detecting the presence of the object is based on detecting a disturbance in a magnetic field within the connection area that is indicative of a presence of a ferromagnetic material of the connection structure within the connection area, and wherein detecting the absence of the object is based on detecting a disturbance in the magnetic field within the connection area that is indicative of an absence of the ferromagnetic material of the connection structure within the connection area.

(40) Clause 18. The method according to clause 16 or 17, wherein performing the at least one action includes providing at least one of a visual indication, an audio indication, and a tactile indication based on the connection status.

(41) Clause 19. The method according to any of clauses 16-18, further comprising: receiving, with the control device, communication data gathered by at least one short-range wireless communication antenna configured for detecting a presence or an absence of an identification element on the connection structure, wherein determining the connection status of the connector is further based on detecting the presence or the absence of the identification element on the connection structure.

(42) Clause 20. A connector for a fall protection compliance system, the connector comprising: a frame having an opening and a connection area configured for receiving a connection structure therein; and a connection sensor assembly associated with the frame, the connection sensor assembly comprising: at least one permanent magnet configured for generating a magnetic field within at least a portion of the connection area; at least one magnetometer configured for detecting a presence or an absence of a disturbance in the magnetic field; at least one short-range wireless communication antenna configured for detecting a presence or an absence of an identification element on the connection structure; and a control device comprising at least one processor programmed or configured to: receive connection data from the at least one magnetometer and communication data from the at least one short-range wireless communication antenna; determine a connection status of the connector based on detecting the presence or the absence of the disturbance in the magnetic field and based on detecting the presence or the absence of the identification element; and perform at least one action based on the connection status, wherein detecting the presence of the disturbance in the magnetic field is indicative of a presence of a ferromagnetic material of the connection structure within the connection area, wherein detecting the absence of the disturbance in the magnetic field is indicative of an absence of the ferromagnetic material of the connection structure within the connection area, wherein detecting the presence of the identification element is indicative of a presence of an approved connection structure, and wherein detecting the absence of the identification element is indicative of an absence of the approved connection structure.

(43) Clause 21. A connector for a fall protection compliance system, the connector comprising: a frame having an opening and a connection area configured for receiving a connection structure therein; and a connection sensor assembly associated with the frame, the connection sensor assembly comprising: at least one short-range wireless communication antenna configured for detecting a presence or an absence of an identification element on the connection structure; and a control device comprising at least one processor programmed or configured to: receive

communication data from the at least one short-range wireless communication antenna; determine a connection status of the connector based on detecting the presence or the absence of the identification element; and perform at least one action based on the connection status.

(44) Clause 22. The connector according to clause 21, wherein detecting the presence of the identification element is indicative of a presence of an approved connection structure, and wherein detecting the absence of the identification element is indicative of an absence of the approved connection structure.

(45) Clause 23. The connector according to clause 21 or 22, wherein the connection status comprises a connected status indicative of a presence of an approved connection structure, and a disconnected status indicative of an absence of the approved connection structure.

(46) Clause 24. The connector according to any of clauses 21-23, wherein the at least one action includes providing at least one of a visual indication, an audio indication, and a tactile indication based on the connection status.

(47) Clause 25. The connector according to any of clauses 21-24, wherein the connection sensor assembly further comprises a communication interface, and wherein the at least one action includes transmitting the connection status to a remote device using the communication interface.

(48) These and other features and characteristics of the present disclosure, as well as the methods of operation and functions of the related elements of structures and the combination of parts and economies of manufacture, will become more apparent upon consideration of the following description and the appended claims with reference to the accompanying drawings, all of which form a part of this specification, wherein like reference numerals designate corresponding parts in the various figures. It is to be expressly understood, however, that the drawings are for the purpose of illustration and description only and are not intended as a definition of the limits of the disclosure. Hence, specific dimensions and other physical characteristics related to the embodiments disclosed herein are not to be considered as limiting. Further, it is to be understood that the disclosure may assume various alternative variations and step sequences, except where expressly specified to the contrary.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Additional advantages and details of non-limiting embodiments or aspects are explained in greater detail below with reference to the exemplary embodiments that are illustrated in the accompanying schematic figures, in which:
- (2) FIG. 1 is an illustration of a fall protection compliance system in accordance with some non-limiting embodiments or aspects of the present disclosure;
- (3) FIG. 2 is a top view of fall protection safety equipment having a connector in accordance with some non-limiting embodiments or aspects of the present disclosure;
- (4) FIG. 3 is a perspective view of one of the connectors shown in FIG. 2 connected to a connection structure;
- (5) FIG. 4 is a side view of the connector and connection structure shown in FIG. 3;
- (6) FIG. 5 is an exploded perspective view of the connector shown in FIG. 4;
- (7) FIG. 6A is a perspective view of a connection structure in accordance with some non-limiting embodiments or aspects of the present disclosure;
- (8) FIG. 6B is a perspective view of a connection structure in accordance with some non-limiting embodiments or aspects of the present disclosure;
- (9) FIG. 7 is an illustration of a fall protection compliance system in accordance with some non-limiting embodiments or aspects of the present disclosure;
- (10) FIG. 8 is an illustration of a fall protection compliance system in accordance with some non-

limiting embodiments or aspects of the present disclosure;

(11) FIG. 9 is a diagram of some non-limiting embodiments or aspects of a fall protection compliance system;

(12) FIG. 10 is a diagram of components of a connection sensor assembly in accordance with some non-limiting embodiments or aspects of the present disclosure;

(13) FIG. 11 is an illustration of a graphical user interface for use with a fall protection compliance system in accordance with some non-limiting embodiments or aspects of the present disclosure;

(14) FIG. 12 is an illustration of a graphical user interface for use with a fall protection compliance system in accordance with some non-limiting embodiments or aspects of the present disclosure;

and

(15) FIG. 13 is a flowchart of a process for detecting connection of a connector of a fall protection compliance system to a connection structure in accordance with some non-limiting embodiments or aspects of the present disclosure.

(16) FIG. 14 is a diagram depicting a hook connection structure having a plurality of sensors provided thereon.

(17) FIG. 15 is a diagram depicting an example configuration of an interlock switch configured to monitor a position of a movable gate of a hook connection structure.

(18) FIG. 16 is a diagram depicting an example configuration of a piezo sensor and a pressure switch.

(19) FIG. 17 is a diagram depicting an example configuration of a piezo sensor and a pressure switch.

(20) FIG. 18 is a diagram depicting example piezo sensor outputs when a connection structure is provided in different operating conditions.

(21) FIG. 19 is a graph depicting a signature associated with a connected user moving relative to a horizontal support line anchor structure.

(22) FIG. 20 is a diagram depicting an example piezo sensor output indicative of a connected user.

(23) FIG. 21 is a flow diagram depicting an example method of providing fall protection compliance monitoring.

(24) FIG. 22 is a flow diagram depicting another example method for providing fall protection compliance monitoring.

(25) FIGS. 23A-H depict examples of gate structure monitoring sensors.

(26) FIG. 24 is a perspective view of one of the connectors shown in FIG. 2 connected to a connection structure shown in FIG. 25

(27) FIG. 25 is a diagram depicting a connection structure with a rotatable housing in accordance with some non-limiting embodiments or aspects of the present disclosure.

(28) FIG. 26 is a diagram depicting the first and second component of the rotatable housing shown in FIG. 24.

(29) FIGS. 27A-D are a front, bottom, side and perspective view of the first component of the rotatable housing.

(30) FIGS. 28A-D are a front, bottom, side and perspective view of the second component of the rotatable housing.

(31) FIGS. 29A-B are diagrams showing the snaps in the first component.

(32) FIGS. 29C-D are diagrams showing the hooks in the second component.

(33) FIG. 30 is a diagram depicting the connection structure with and without the rotatable housing.

(34) FIG. 31A-B are diagrams showing the rotatable housing moving with the connector from Position I to Position II.

(35) It should be appreciated by those skilled in the art that any block diagrams herein represent conceptual views of illustrative systems embodying the principles of the present subject matter. Similarly, it will be appreciated that any flow charts, flow diagrams, state transition diagrams, pseudo code, and the like represent various processes which may be substantially represented in

computer-readable medium and executed by a computer or processor, whether such computer or processor is explicitly shown. While each of the figures illustrates a particular embodiment for purposes of illustrating a clear example, other embodiments may omit, add to, reorder, and/or modify any of the elements shown in the figures.

DETAILED DESCRIPTION

(36) For purposes of the description hereinafter, the terms “end”, “upper”, “lower”, “right”, “left”, “vertical”, “horizontal”, “top”, “bottom”, “lateral”, “longitudinal” and derivatives thereof shall relate to the disclosure as it is oriented in the drawing figures. However, it is to be understood that the disclosure may assume various alternative variations and step sequences, except where expressly specified to the contrary.

(37) All numbers and ranges used in the specification and claims are to be understood as being modified in all instances by the term “about”. By “about” is meant plus or minus twenty-five percent of the stated value, such as plus or minus ten percent of the stated value. However, this should not be considered as limiting to any analysis of the values under the doctrine of equivalents.

(38) Unless otherwise indicated, all ranges or ratios disclosed herein are to be understood to encompass the beginning and ending values and any and all subranges or subratios subsumed therein. For example, a stated range or ratio of “1 to 10” should be considered to include any and all subranges or subratios between (and inclusive of) the minimum value of 1 and the maximum value of 10; that is, all subranges or subratios beginning with a minimum value of 1 or more and ending with a maximum value of 10 or less. The ranges and/or ratios disclosed herein represent the average values over the specified range and/or ratio.

(39) The terms “first”, “second”, and the like are not intended to refer to any particular order or chronology, but refer to different conditions, properties, or elements.

(40) The term “at least” is synonymous with “greater than or equal to”.

(41) As used herein, “at least one of” is synonymous with “one or more of”. For example, the phrase “at least one of A, B, and C” means any one of A, B, or C, or any combination of any two or more of A, B, or C. For example, “at least one of A, B, and C” includes one or more of A alone; or one or more B alone; or one or more of C alone; or one or more of A and one or more of B; or one or more of A and one or more of C; or one or more of B and one or more of C; or one or more of all of A, B, and C.

(42) As used herein, the terms “parallel” or “substantially parallel” mean a relative angle as between two objects (if extended to theoretical intersection), such as elongated objects and including reference lines, that is from 0° to 5°, or from 0° to 3°, or from 0° to 2°, or from 0° to 1°, or from 0° to 0.5°, or from 0° to 0.25°, or from 0° to 0.1°, inclusive of the recited values.

(43) As used herein, the terms “perpendicular” or “substantially perpendicular” mean a relative angle as between two objects at their real or theoretical intersection is from 85° to 90°, or from 87° to 90°, or from 88° to 90°, or from 89° to 90°, or from 89.5° to 90°, or from 89.75° to 90°, or from 89.9° to 90°, inclusive of the recited values.

(44) In the present document, the word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment or implementation of the present subject matter described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments.

(45) The terms “comprises”, “comprising”, or any other variations thereof, are intended to cover a non-exclusive inclusion, such that a setup, device, or method that comprises a list of components or steps does not include only those components or steps but may include other components or steps not expressly listed or inherent to such setup, device, or method. In other words, one or more elements in a system or apparatus preceded by “comprises . . . a” does not, without more constraints, preclude the existence of other elements or additional elements in the system or method.

(46) The terms “includes”, “including”, or any other variations thereof are intended to cover a non-

exclusive inclusion such that a setup, device, or method that includes a list of components or steps does not include only those components or steps but may include other components or steps not expressly listed or inherent to such setup, device, or method. In other words, one or more elements in a system or apparatus preceded by “includes . . . a” does not, without more constraints, preclude the existence of other elements or additional elements in the system or method.

(47) The terms “an embodiment”, “embodiment”, “embodiments”, “the embodiment”, “the embodiments”, “one or more embodiments”, “some non-limiting embodiments or aspects”, and “one embodiment” mean “one or more (but not all) embodiments of the invention(s)” unless expressly specified otherwise. A description of an embodiment with several components in communication with each other does not imply that all such components are required. On the contrary, a variety of optional components is described to illustrate the wide variety of possible embodiments of the disclosure.

(48) No aspect, component, element, structure, act, step, function, instruction, and/or the like used herein should be construed as critical or essential unless explicitly described as such. Also, as used herein, the articles “a” and “an” are intended to include one or more items and may be used interchangeably with “one or more” and “at least one.” Furthermore, as used herein, the term “set” is intended to include one or more items (e.g., related items, unrelated items, a combination of related and unrelated items, and/or the like) and may be used interchangeably with “one or more” or “at least one.” Where only one item is intended, the term “one” or similar language is used. Also, as used herein, the terms “has”, “have”, “having”, or the like are intended to be open-ended terms. Further, the phrase “based on” is intended to mean “based at least in partially on” unless explicitly stated otherwise. The term “some non-limiting embodiments or aspects” means “one or more (but not all) embodiments or aspects of the disclosure(s)” unless expressly specified otherwise. A description of some non-limiting embodiments or aspects with several components in communication with each other does not imply that all such components are required. On the contrary, a variety of optional components is described to illustrate the wide variety of possible embodiments of the disclosure.

(49) When a single device or article is described herein, it will be clear that more than one device/article (whether they cooperate) may be used in place of a single device/article. Similarly, where more than one device or article is described herein (whether they cooperate), it will be clear that a single device/article may be used in place of the more than one device or article or a different number of devices/articles may be used instead of the shown number of devices or programs. The functionality and/or the features of a device may be alternatively embodied by one or more other devices which are not explicitly described as having such functionality/features. Thus, other embodiments of the disclosure need not include the device itself.

(50) As used herein, the terms “communication”, “communicate”, “send”, and/or “receive” may refer to the reception, receipt, transmission, transfer, provision, and/or the like of information (e.g., data, signals, messages, instructions, commands, and/or the like). For one unit (e.g., a device, a system, a component of a device or system, combinations thereof, and/or the like) to be in communication with another unit means that the one unit is able to directly or indirectly receive information from and/or transmit information to the other unit. This may refer to a direct or indirect connection (e.g., a direct communication connection, an indirect communication connection, and/or the like) that is wired and/or wireless in nature. Additionally, two units may be in communication with each other even though the information transmitted may be modified, processed, relayed, and/or routed between the first and second unit. For example, a first unit may be in communication with a second unit even though the first unit passively receives information and does not actively transmit information to the second unit. As another example, a first unit may be in communication with a second unit if at least one intermediary unit (e.g., a third unit located between the first unit and the second unit) processes information received from the first unit and communicates the processed information to the second unit. In some non-limiting embodiments or aspects, a message

may refer to a network packet (e.g., a data packet and/or the like) that includes data. It will be appreciated that numerous other arrangements are possible.

(51) As used herein, the terms “server” and/or “processor” may refer to one or more computing devices, such as processors, storage devices, and/or similar computer components that communicate with client devices and/or other computing devices over a network, such as the Internet or private networks, and, in some examples, facilitate communication among other servers and/or client devices. It will be appreciated that various other arrangements are possible. As used herein, the term “system” may refer to one or more computing devices or combinations of computing devices such as, but not limited to, processors, servers, client devices, software applications, and/or other like components. In addition, reference to “a server” or “a processor”, as used herein, may refer to a previously-recited server and/or processor that is recited as performing a previous step or function, a different server and/or processor, and/or a combination of servers and/or processors. For example, as used in the specification and the claims, a first server and/or a first processor that is recited as performing a first step or function may refer to the same or different server and/or a processor recited as performing a second step or function.

(52) As used herein, the term “remote device” may refer to one or more computing devices, which may be used by a remote user, such as an industrial hygienist or a project manager, to monitor compliant use of a fall protection compliance system. In some non-limiting embodiments, a remote device may include a computing device configured to communicate with one or more networks and/or facilitate at least one of receiving and sending information from and to a connector, such as, but not limited to, one or more desktop computers, one or more mobile devices, and/or other like devices.

(53) As used herein, the term “computing device” may refer to one or more electronic devices that are configured to directly or indirectly communicate with or over one or more networks. In some non-limiting embodiments, a computing device may include a mobile device. A mobile device may include a smartphone, a portable computer, a wearable device (e.g., watches, glasses, lenses, clothing, and/or the like), a personal digital assistant (PDA), and/or other like devices. In some non-limiting embodiments, a computing device may include a server, a desktop computer, and/or the like.

(54) As used herein, the term “system” may refer to one or more computing devices or combinations of computing devices such as, but not limited to, processors, servers, client devices, software applications, and/or other like components. In addition, reference to “a server” or “a processor,” as used herein, may refer to a previously-recited server and/or processor that is recited as performing a previous step or function, a different server and/or processor, and/or a combination of servers and/or processors. For example, as used in the specification and the claims, a first server and/or a first processor that is recited as performing a first step or function may refer to the same or different server and/or a processor recited as performing a second step or function.

(55) As discussed herein, certain operations may be performed in a different order, modified, or removed. Moreover, steps may be added to the above-described logic and still conform to the described embodiments. Further, operations described herein may occur sequentially or certain operations may be processed in parallel. Yet further, operations may be performed by a single processing unit or by distributed processing units.

(56) In the following detailed description of the embodiments of the disclosure, reference is made to the accompanying drawings that form a part hereof, and in which are shown by way of illustration specific embodiments in which the disclosure may be practiced. It should be understood, however, that it is not intended to limit the disclosure to the forms disclosed, but on the contrary, the disclosure is to cover all modifications, equivalents, and alternatives falling within the spirit and the scope of the disclosure. It is to be understood that other embodiments may be utilized and that changes may be made without departing from the scope of the present disclosure. The following description is, therefore, not to be taken in a limiting sense.

(57) Various embodiments or aspects of the present disclosure are directed to comprehensive fall protection compliance systems and methods for directly monitoring worker safety while using safety equipment, such as fall protection equipment. In some non-limiting embodiments or aspects, the system may be a passive monitoring system that collects data regarding the direct usage and coupling of fall protection equipment, such as, without limitation, harnesses, lanyards, anchorages, horizontal and vertical lifelines, as well as winches, davits, and other raising and lowering equipment.

(58) In some non-limiting embodiments or aspects, the fall protection compliance system may be configured to detect, such as using one or more electronic sensors, that a connection has been made between a user wearing a fall protection harness or similar safety equipment, and an anchor point. The system may be further configured to indicate the proper connection, disconnection, and usage compliance of the safety equipment regarding an unsafe or safe condition or area. The system may be configured to monitor the connection between the user and the safety equipment and determine whether the worker is securely connected to the safety equipment. In some non-limiting embodiments or aspects, the system may be configured to store information regarding various aspects of the system, such as proper connection of the safety equipment, length of use of the safety equipment, identification of connection to a particular piece of safety equipment. The stored data may have a time and date stamp. In some non-limiting embodiments or aspects, the stored data can be transmitted to a remote device, such as a cell phone or an external monitoring station, and/or stored locally in memory storage for later retrieval.

(59) In some non-limiting embodiments or aspects, the stored data may indicate the worker identification number, the device identification number, current time, total time that the worker is wearing or connected to the safety equipment, and individual periods of use or non-use of the safety equipment. The system may have built-in algorithms and safeties to both indicate that the user is using the equipment in compliance with standards, but also to prevent tampering or obfuscation of the results. This ensures that the user uses the equipment properly and eliminates the need for direct physical inspection of compliance by an industrial hygienist or compliance officer.

(60) With initial reference to FIG. 1, a fall protection compliance system **100** is illustrated in accordance with some non-limiting embodiments or aspects of the present disclosure. The fall protection compliance system **100** includes at least one piece of fall protection safety equipment **102** that is configured for securing a user U wearing a harness **104** to an anchor **106** secured to a wall W, floor F, ceiling, or other component of a structure S. In some non-limiting embodiments or aspects, the fall protection safety equipment **102** may have a first connector **200a** at its first end and a second connector **200b** at its second end. The first connector **200a** may be configured for connecting directly to the harness **104** or another piece of safety equipment, such as an energy absorber, that is directly connected to the harness **104**. The second connector **200b** may be configured for connecting directly to the anchor **106** or to another piece of safety equipment, such as an energy absorber, that is connected directly to the anchor **106**. In this manner, the fall protection safety equipment **102** is configured to connect the user U wearing the harness **104** to the anchor **106**. The fall protection safety equipment **102**, the harness **104**, and the anchor **106** together define the fall protection compliance system **100**.

(61) In some non-limiting embodiments or aspects, the fall protection safety equipment **102** may be a lanyard **110** having the first connector **200a** at its first end and the second connector **200b** at its second end. In other non-limiting embodiments or aspects, the fall protection safety equipment **102** may be a line retraction device, such as a self-retracting lanyard (SRL). The SRL may have a safety line that is configured to be unwound (paid out) from a drum when a certain level of tension is applied to the safety line, such as during movement of the user U on the structure S. When such tension is reduced or released, the SRL is configured to slowly rotate in a reverse direction, thereby causing the safety line to retract or rewind onto the drum.

(62) As further described herein, the fall protection safety equipment **102** includes one or more

sensors and associated control devices configured to gather data in real-time as the user U engages in activities on the structure S while wearing the fall protection safety equipment **102**. For example, the fall protection safety equipment **102** may include one or more sensors configured to detect a connection status of at least one of the first and second connectors **200a**, **200b**. In addition, the fall protection safety equipment **102** may include one or more output devices for outputting data that is indicative of the connection status of at least one of the first and second connectors **200a**, **200b**. For example, the fall protection safety equipment **102** may include one or more devices to generate at least one of an audible feedback (e.g., one or more speakers), a visual feedback (e.g., one or more displays, light emitting diodes (LEDs), or the like), or a tactile feedback (e.g., a vibration device). In addition, the fall protection safety equipment **102** may be configured to transmit the connection status of at least one of the first and second connectors **200a**, **200b** to a remote device.

(63) With reference to FIG. 2, the fall protection safety equipment **102** is shown separate from the fall protection compliance system **100**. The fall protection safety equipment **102** has the first connector **200a** connected to a first end **112a** of the lanyard **110** and the second connector **200b** connected to a second end **112b** of the lanyard **110**. In some non-limiting embodiments or aspects, the fall protection safety equipment **102** may have a third connector **200c** connected to a separate line **114** that is connected to the lanyard **110** between the first end **112a** and the second end **112b**. As discussed herein, the first connector **200a** may be configured for connecting directly to the harness **104** (shown in FIG. 1) or another piece of safety equipment, such as an energy absorber, that is directly connected to the harness **104**. The second connector **200b** may be configured for connecting directly to the anchor **106** (shown in FIG. 1) or to another piece of safety equipment, such as an energy absorber, that is connected directly to the anchor **106**. The third connector **200c** may be configured for connecting to a second anchor **106** on the structure S. The third connector **200c** may be connected to the second anchor **106** before the second connector **200b** is removed from the first anchor **106** to allow the user U to move on the structure S while remaining connected to at least one of the anchors **106**. As described herein, at least one of the connectors **200a**, **200b**, **200c** may have one or more sensors configured to detect a connection status that is indicative of whether at least one of the connectors **200a**, **200b**, **200c** is connected to the connection structure, such as the harness **104**, the anchor **106**, or other connection structure.

(64) With reference to FIGS. 3-4, the connector **200** and a connection structure **300** are illustrated in accordance with some non-limiting embodiments or examples of the present disclosure. While FIG. 3 illustrates the connector **200** that is configured as a snap hook, in other non-limiting embodiments or aspects, the connector **200** may be a carabiner or the like. Similarly, while the connection structure **300** is shown as a D-ring, in other non-limiting embodiments or aspects, the connection structure **300** may be a U-bolt or the like. It should be understood that the structures and techniques described herein can be applied to a variety of other devices configured for securing a user U to an anchor **106** as part of a fall protection compliance system **100**.

(65) In some non-limiting embodiments or aspects, the connector **200** can be configured for use as an additional component to existing safety equipment, such as by being placed intermediate between the harness **104** and the fall protection safety equipment **102** (shown in FIG. 1), and/or between the anchor **106** and the fall protection safety equipment **102**. In such embodiments or aspects, the connector **200** may be positioned intermediate between the D-ring of the harness **104** and the lanyard **110** or other fall protection device **102**. In this configuration, the connector **200** may have two connection points to allow connection between the harness **104** and the lanyard **110** such that at least a portion of the connector **200** functions as a load bearing member.

(66) With continued reference to FIGS. 3-4, the connector **200** has a frame **202** configured for connecting the connector **200** to at least a portion of at least one piece of safety equipment, such as the connection structure **300**. The frame **202** may have a connection point **201** for connecting the lanyard **110** (shown in FIG. 1) or other piece of fall protection safety equipment to the connector **200**. The frame **202** may have a substantially C-shaped structure with an opening **208** extending

between two opposed ends of the C-shaped frame **202**.

(67) With continued reference to FIGS. **3-4**, the connector **200** has a movable gate **204** connected to the frame **202** and movable between a closed position and an open position. In the closed position, such as shown in FIG. **3**, the movable gate **204** encloses a connection area **206** to prevent passage of the connection structure **300** or other object through the opening **208** on the frame **202**. The movable gate **204** contacts the frame **202** such that a continuous loop encloses the connection area **206** and prevents movement of the connecting structure **300** or other objection into or from the connection area **206**. The movable gate **204** may be pivotally or slidably movable to the open position, such as by rotating about a pivot point **210** in a direction of arrow A. With the movable gate **204** in the open position, the opening **208** on the frame **202** is unobstructed to allow passage of the connection structure **300** or other object into and out of the connection area **206**.

(68) In some non-limiting embodiments or aspects, the movable gate **204** may be biased to the closed position. For example, a biasing member, such as a spring **203**, may be provided to bias the movable gate **204** to the closed position. In some non-limiting embodiments or aspects, one or more sensors may be provided for detecting the open and/or closed position of the movable gate **204**, and/or a movement of the movable gate **204** toward or away from the open/closed position. The movable gate **204** may have a locking assembly **205** for maintaining the movable gate **204** in the closed position and/or permitting movement of the movable gate **204** in a direction from the closed position toward the open position. The locking assembly **205** may be operatively connected to the gate **204** and movable between a first position and a second position, wherein, in the first position, the locking assembly **205** is configured for preventing movement of the gate **204** from the closed position toward the open position, and wherein, in the second position, the locking assembly **205** is configured for permitting movement of the gate **204** from the closed position toward the open position. For example, movement of the movable gate **204** from the closed position may be prevented unless the locking assembly **205** is disengaged (i.e., moved from the first position toward the second position). In this manner, inadvertent opening of the movable gate **204** can be prevented. The locking assembly **205** may be connected to the movable gate **204** by way of a channel **207** that slidably receives a pin **209** of the movable gate **204**. The locking assembly **205** may be pivotally movable about a second pivot point **211**. When a user operates the locking assembly **205** (e.g., the user squeezes the locking assembly **205** against the frame **202**), the locking assembly **205** pivotally moves about the second pivot point **211** to allow the pin **209** to move within the channel **207**, which thereby permits movement of the movable gate **204** from the closed position toward the open position.

(69) With reference to FIG. **5**, and with continued reference to FIGS. **3-4**, the connector **200** has a connection sensor assembly **214** configured for determining a connection status of the connector **200**. A connection status generally refers to a determination, using the connection sensor assembly **214**, as to whether the connector **200** is connected to a connection structure **300** or disconnected from the connection structure **300**. For example, the connection status may indicate that the connector **200** is connected to the connection structure **300**, such as when the connection sensor assembly **214** detects the presence of a ferromagnetic material of the connection structure within the connection area **206** and/or detects a presence of an identification element associated with the connection structure **300**, as described herein. Alternatively, the connection status may indicate that the connector **200** is disconnected from the connection structure **300**, such as when the connection sensor assembly **214** detects an absence of a ferromagnetic material within the connection area **206** and/or detects an absence of an identification element **320** associated with the connection structure **300**, as described herein, as described herein.

(70) With continued reference to FIG. **5**, the connection sensor assembly **214** is enclosed within a housing **213** connected to the frame **202** of the connector **200**. In some non-limiting embodiments or aspects, the housing **213** may have a pair of housing sides that connect to each other to enclose the components of the connection sensor assembly **214**. The connection sensor assembly **214** may

be configured to detect whether the connector **200** is connected to safety equipment, such as a component of a fall protection compliance system **100**. In some non-limiting embodiments or aspects, the connection sensor assembly **214** may be further configured to detect a presence of an identification element associated with the connection structure **300**. In some non-limiting embodiments or aspects, the connection sensor assembly **214** may be further configured to detect movement of the connector **200**, such as due to movement of the user **U** to which the connector **200** is directly or indirectly attached. The connection sensor assembly **214** may have one or more sensors, including one or more accelerometers, gyroscopes, pressure sensors, magnetic sensors, contact switches, RFID, NFC, and other radio-based proximity detectors, as discussed herein.

(71) With continued reference to FIG. 5, the connection sensor assembly **214** has one or more sensors **215** and a control device **216** operatively connected to the one or more sensors **215** to receive data from the one or more sensors **215** regarding the connection status of the connector **200**. The plurality of sensors **215** may be disposed on a side of the frame **202** or on an inner surface **212** facing the connection area **206**. In some non-limiting embodiments or aspects, the inner surface **212** may be a surface of the frame **202** that is configured for contacting the connection structure **300** (shown in FIG. 4) when the connector **200** is connected to the connection structure **300**.

(72) In some non-limiting embodiments or aspects, the one or more sensors **215** may include at least one magnetometer **217**. In some non-limiting embodiments or aspects, a plurality of magnetometers **217** may be spaced apart in an arcuate arrangement that corresponds to an arcuate shape of the frame **202**. The magnetometers **217** may be disposed in an area of the frame **202** that corresponds to a probable location where the connection structure **300** will be located when the connection structure **300** is disposed within the connection area **206** of the connector **200**. At least one permanent magnet, such as a rare earth magnet **219**, may be associated with at least one magnetometer **217**, such as by being disposed between a plurality of magnetometers **217**. The rare earth magnet **219** generates a pre-defined magnetic field within at least a portion of the connection area **206**. When a ferromagnetic material, such as the ferromagnetic metal material of the connection structure **300** or other ferromagnetic object, is inserted into the connection area **206**, the pre-defined magnetic field generated by the rare earth magnet **219** is disturbed by the presence of such a ferromagnetic material. By measuring the disturbance of the pre-defined magnetic field, the connection sensor assembly **214** is configured to detect the connection status of the connector **200**, such as whether the connection structure **300** is disposed in the connection area **206** of the connector **200**.

(73) With continued reference to FIG. 5, the one or more sensors **215** may include at least one short-range wireless communication antenna **221** configured for detecting a presence or an absence of an identification element **320** on the connection structure **300**. In some non-limiting embodiments or aspects, the at least one short-range wireless communication antenna **221** may be an RFID antenna and the identification element **320** may be an RFID tag. In some non-limiting embodiments or aspects, the at least one short-range wireless communication antenna **221** may be configured to interact with the identification element **320** (shown in FIGS. 6A-6B) attached to the connection structure **300**. In this manner, when the connection structure **300** having the identification element **320** is connected to the connector **200**, the at least one short-range wireless communication antenna **221** detects the presence of the identification tag **320** on the connection structure **300**, thereby indicating that the connector **300** is connected to an approved connection structure **300**. The at least one short-range wireless communication antenna **221** may be configured as a flex circuit positioned on an upper arcuate portion of the housing **213**. In this manner, the at least one short-range wireless communication antenna **221** is positioned closest to a probable location of the identification element **320** when the connection structure **300** is disposed within the connection area **206** of the connector **200**.

(74) In various non-limiting embodiments or aspects, sensor detection intervals can be optimized

with low duty cycles in order to extend battery life. In this manner, the one or more sensors **215** can be turned on, come to steady state, take a record or data point, and then turn off using a particular cycle. If, for example, one or more sensors **215** can complete its entire cycle within only 100 milliseconds, the connector **200** may have a duty cycle of 1% while generating a data point once every 10 seconds. Depending on the type of sensor used for any of the data acquisition requirements, the sensor data acquisition cycle could be longer or shorter.

(75) The connection sensor assembly **214** may be in a standby or sleep mode prior to connection of the connection structure **300** in order to conserve battery life. In some non-limiting embodiments or aspects, the connection sensor assembly **214** may be activated from the standby or sleep mode by movement of the movable gate **204** and/or the locking assembly **205** from the closed position to the open position. In some non-limiting embodiments or aspects, the connection sensor assembly **214** may be activated from the standby or sleep mode by detecting movement of the connector **200**, such as using a gyroscope or an accelerometer. In some non-limiting embodiments or aspects, the connection sensor assembly **214** may be activated from the standby or sleep mode after an initial connection to a connection structure **300** is made. In some non-limiting embodiments or aspects, the connection sensor assembly **214** may be activated from the standby mode by pressing a button, such as a button on the housing **213** of the connector **200**. In some non-limiting embodiments or aspects, the connection sensor assembly **214** may be activated from the standby or sleep mode by the user being physically present within a work zone where operation of the connector **200** is desired. For example, the zone may have weight sensors and/or light bar sensors that, once activated due to the user's weight or due to the light beam being broken by the user, an activation signal is sent to the connection sensor assembly **214**. The connection sensor assembly **214** may be activated if the user is detected within a predetermined distance of a connection structure where the use of the connector **200** may be required. In further examples, the user may scan into the work area, such as using an RFID tag, and/or the user's presence may be sensed, for example using a sonar or other sensing device, in order to activate the connection sensor assembly **214**. In some non-limiting embodiments or aspects, the connection sensor assembly **214** may be activated from the standby or sleep mode when the connector **200** is unplugged from a power source or a home station, such as when the connector **200** is being recharged. Plugging the connector back to the power source or the home station may cause the connection sensor assembly **214** to enter the standby or sleep mode.

(76) Once activated from the standby or sleep mode, the connection sensor assembly **214** may be configured to detect whether a ferromagnetic material, such as the ferromagnetic material of the connection structure **300** is disposed within the connection area **206** of the connector **200** and/or to detect the presence or absence of an identification element associated with the connection structure **300**.

(77) In some non-limiting embodiments or aspects, the connector **200** may have a communication interface **220** for communicating information regarding the connection status of the connector **200**. In some non-limiting embodiments or aspects, the communication interface **220** may be at least one of a visual, audio, vibration, and tactile indicator on the connector **200** configured for indicating the connection status of the connector **200**. For example, the communication interface **220** may be one or more lights **240** that indicate a connected or disconnected status of the connector. The one or more lights **240** may have a first visual indication, such as a green light, indicative of a presence of a ferromagnetic material of the connection structure **300** within the connection area **206** and/or a presence of an identification element **320** on the connection structure **300**, and a second visual indication, such as a red light, indicative of an absence of the ferromagnetic material of the connection structure **300** within the connection area **206** and/or an absence of an identification element **320** on the connection structure **300**. The first visual indication may be the same or different from the second visual indication.

(78) In some non-limiting embodiments or aspects, the communication interface **220** may be one or

more speakers **242** or other audio devices configured to provide a first audio indication, such as a first tone, indicative of a presence of a ferromagnetic material of the connection structure **300** within the connection area **206** and/or a presence of an identification element **320** on the connection structure **300**, and a second audio indication, such as a second tone different from the first tone, indicative of an absence of the ferromagnetic material of the connection structure **300** within the connection area **206** and/or an absence of an identification element **320** on the connection structure **300**. The first audio indication may be the same or different from the second audio indication.

(79) In some non-limiting embodiments or aspects, the communication interface **220** may be one or more vibration devices configured to provide a first tactile indication, such as a first vibration, indicative of a presence of a ferromagnetic material of the connection structure **300** within the connection area **206** and/or a presence of an identification element **320** on the connection structure **300**, and a second tactile indication, such as a second vibration different from the first vibration, indicative of an absence of the ferromagnetic material of the connection structure **300** within the connection area **206** and/or an absence of an identification element **320** on the connection structure **300**. The first tactile indication may be the same or different from the second tactile indication.

(80) The communication interface **220** may include a transceiver-like component (e.g., a transceiver, a separate receiver and transmitter, etc.) that enables the control device **216** to communicate with other devices, such as via a wired connection, a wireless connection, or a combination of wired and wireless connections. The communication interface **220** may permit the control device **216** to receive information from another device and/or provide information to another device. For example, the communication interface **220** may include an Ethernet interface, an optical interface, a coaxial interface, an infrared interface, a radio frequency (RF) interface, a universal serial bus (USB) interface, a Wi-Fi® interface, low-energy Bluetooth® interface, a cellular network interface, and/or the like.

(81) At any given change of connection status of the connector **200**, a notification may be generated and communicated to the user via the communication interface **220** regarding the current connection state of the connector **200**. These notifications can be segregated into several different categories. For example a green light and a short beep may indicate that the connector **200** is connected to an approved connection structure **300** (i.e., a ferromagnetic connection structure **300** having an identification tag **320**), while a long tone and a red light may be used to indicate that the connector **200** is not connected to the connection structure **300** and/or that it is connected to a ferromagnetic connection structure **300** that does not have an identification element **320**. Other notifications can be generated using various combinations of communication devices associated with the communication interface **220**.

(82) In some non-limiting embodiments or aspects, the communication interface **220** may have a transmitter **222** for wirelessly transmitting data regarding the status of the connector **200** to a remote device **250** (shown in FIG. **10**), such as a cell phone or a computer terminal, and/or receiving data from the remote device **250** using a wireless communication protocol. Each of the one or more sensors **215** is configured to communicate data, such as sensed motions, events and conditions, via wireless communications. The transmitter **222** may be a short range or long range radio, low-energy Bluetooth® (BLE), near-field communication (NFC), or other transmitter configured to communicate data from the connector **200** continuously or periodically. The connector **200** can be interrogated using short range radio communications, standard networking protocols, direct connection of a port, or the charging connection.

(83) Data recorded by the connector **200** may be transmitted for download by the remote device **250**. The transmitted data may include, for example, the time-stamped transitions between all device states, sensor data, and device health status. Once data is downloaded, it can be reviewed by the compliance manager, industrial hygienist or other supervisor personnel. Data may be password protected so that it cannot be overwritten or deleted without being recorded to a permanent record.

(84) In some non-limiting embodiments or aspects, time and date stamps, along with the unit identification number can be transmitted to the remote device **250**. In this manner, if the user is in an area where full compliance is expected (both conditions met), but at least one of the conditions is not met, a safety officer can be dispatched to physically inspect the user to determine the cause for non-compliance. In the event of a fall, the connector **200** can be configured to transmit this event to the remote device **250**. In some non-limiting embodiments or aspects, the connector **200** may be configured to transmit a distress signal, such as by placing a call to emergency services, in an event that the user is in distress based on the data received from the one or more sensors **215**. In some non-limiting embodiments or aspects, the user can activate the connector **200**, such as by pushing a button thereon, to transmit an alert that the user requires assistance, such as in an emergency situation.

(85) The connector **200** may be configured to save data having predetermined characteristics, such as when sensor readings exceed a predetermined threshold, or when the one or more sensors **215** detect or fail to detect a condition. In some non-limiting embodiments or aspects, the connector **200** may be configured to save data when both conditions of compliant use of the connector **200** are met, such as when a proper connection is made. In an event that the connector **200** is disconnected from the connection structure **300**, the connector **200** may store data regarding such event(s).

(86) In some non-limiting embodiments or aspects, the function of determining compliant use of the connector **200** can be performed externally, such as by the application software used by the remote device **250**. In this manner, the information recorded by the connector **200** may be downloaded and evaluated externally to confirm compliance. This function can be carried out on site, remotely using an off-site server or controller, or remotely by reviewing reports submitted by local safety compliance, industrial hygienists, or other safety personnel.

(87) In some non-limiting embodiments or aspects, the connector **200** may be configured for wireless communication prior to issuance to the worker. For example, the connector **200** may be configured to the user, the safety equipment with which the connector **200** is to be used, the work site, area, and/or other data of importance to the safety officer or industrial hygienist. This may be done using a protocol to protect overwriting or editing the data.

(88) With reference to FIG. 5, the control device **216** is operatively connected to the one or more sensors **215** to receive data from the one or more sensors **215** regarding the connection status of the connector **200**. A battery **224** may be provided for powering the components of the connector **200**. In some non-limiting embodiments or aspects, the battery **224** may be a rechargeable lithium polymer battery pack.

(89) The control device **216** may be configured to use data from at least one of the plurality of sensors **215**, such as at least one magnetometer **217** and/or the at least one short-range wireless communication antenna **221**, to determine the connection status of the connector **200**. For example, the control device **216** may determine that the connector **200** has been connected to the connection structure **300** based on a number of ordered operations, such as by receiving data indicating movement of the movable gate **204** from the closed position to the open position. The control device **216** may further receive data from the one or more magnetometers **217** indicating that the connection structure **300** is disposed within the connection area **206**. The control device **216** may further receive data from the at least one short-range wireless communication antenna **221** indicating that the identification element **320** of the connection structure **300** is disposed in proximity of the connector **200**.

(90) After determining that the connector **200** is connected to the connection structure **300**, such as by detecting the presence of the ferromagnetic material of the connection structure **300** within the connection area **206** of the connector **200** and/or detecting the presence of the identification element **320** of the connection structure **300** in proximity of the at least one short-range wireless communication antenna **221** of the connector **200**, the control device **216** may be configured to indicate that a proper connection has been made, such as by generating an audible, visual, or a

tactile alert. For example, if the connector **200** is connected to the connection structure **300**, such as by detecting the presence of the ferromagnetic material of the connection structure **300** within the connection area **206** of the connector **200**, and the at least one short-range wireless communication antenna **221** detects the presence of the identification element **320** that is an expected identification element (i.e., the identification element **320** corresponding to the connection structure **300** to which the connector **200** is supposed to be connected to), the control device **216** may be configured to actuate the communication interface **220** to display a first light (such as a green status light), and/or to emit a first sound, and/or to generate a first tactile response.

(91) If the control device **216** determines that the connector **200** is not connected to the connection structure **300**, such as by failing to detect the presence of the ferromagnetic material of the connection structure **300** within the connection area **206** of the connector **200** and/or failing to detect the presence of the identification element **320** of the connection structure **300** in proximity of the at least one short-range wireless communication antenna **221** of the connector **200**, the control device **216** may be configured to indicate that an improper connection has been made, such as by generating an audible, visual, or a tactile alert. For example, if the connector **200** is not connected to a connection structure or is connected to a ferromagnetic material that is not an approved connection structure **300**, and/or if the at least one short-range wireless communication antenna **221** fails to detect the presence of the identification element **320** or if the at least one short-range wireless communication antenna **221** detects an unexpected identification element (i.e., the identification element **320** corresponding to the connection structure **300** to which the connector **200** is not supposed to be connected to), the control device **216** may be configured to actuate the communication interface **220** to display a second light (such as a red status light), and/or to emit a second sound, and/or to generate a second tactile response, wherein the second light, the second sound, and the second tactile response are different from the first light, the first sound, and the first tactile response, respectively.

(92) With reference to FIGS. **6A-6B**, in some non-limiting embodiments or aspects, the connection structure **300** may be a D-ring connected to an anchor or a harness. The connection structure **300** may be made from a ferromagnetic material and the connector **200** may be configured to detect whether a connection is made with the connection structure **300**, such as by detecting a presence of a magnetic field, as described herein.

(93) With continued reference to FIGS. **6A-6B**, the connection structure **300** has a connection loop **302** that is configured for being received within the connection area **206** of the connector **200** when the movable gate **204** is in the open position. The connection structure **300** further has an anchor point **304** configured as a separate loop for connecting to an anchor. The identification element **320** may be connected to at least a portion of the connection structure **300**. In some non-limiting embodiments or aspects, such as shown in FIG. **6A**, the identification element **320** is connected proximate to the anchor point **304** and away from the connection loop **302** in order to prevent damage to the identification element **320** due to contact with the frame **202** of the connector **200**. In some non-limiting embodiments or aspects, such as shown in FIG. **6B**, the identification element **320** may be provided on a deflectable element **322** that is positioned within the connection loop **302**. In this manner, connection of the connector **200** to the connection loop **302** deflects the deflectable element **322** such that the identification element **320** is positioned proximate to the connection sensor assembly **214**. For example, the identification element **320** may directly contact at least a portion of the connection sensor assembly **214**, such as by contacting the housing **213** of the connection sensor assembly **214**.

(94) Referring now to FIG. **7**, illustrated is a diagram of example components of the control device **216**. As shown in FIG. **7**, control device **216** may include a bus **260**, at least one processor **262**, memory **264**, storage component **268**, input component **270**, output component **272**, and communication interface **220**.

(95) Bus **260** may include a component that permits communication among the components of the

control device **216**. In some non-limiting embodiments or aspects, processor **262** may be implemented in hardware, firmware, or a combination of hardware and software. For example, processor **262** may include a processor (e.g., a central processing unit (CPU), a graphics processing unit (GPU), an accelerated processing unit (APU), etc.), a microprocessor, a digital signal processor (DSP), and/or any processing component (e.g., a field-programmable gate array (FPGA), an application-specific integrated circuit (ASIC), etc.) that can be programmed to perform a function. Memory **264** may include random access memory (RAM), read-only memory (ROM), and/or another type of dynamic or static storage device (e.g., flash memory, magnetic memory, optical memory, etc.) that stores information and/or instructions for use by processor **262**.

(96) The storage component **268** may store information and/or software related to the operation and use of the control device **216**. For example, the storage component **268** may include a hard disk (e.g., a magnetic disk, an optical disk, a magneto-optic disk, a solid state disk, etc.), a compact disc (CD), a digital versatile disc (DVD), a floppy disk, a cartridge, a magnetic tape, and/or another type of computer-readable medium, along with a corresponding drive.

(97) The input component **270** may include a component that permits the control device **216** to receive information, such as via user input (e.g., a touchscreen display, a keyboard, a keypad, a mouse, a button, a switch, a microphone, a camera, etc.). Additionally or alternatively, input component **270** may include a sensor for sensing information (e.g., a global positioning system (GPS) component, an accelerometer, a gyroscope, an actuator, etc.). Output component **272** may include a component that provides output information from the control device **216** (e.g., a display, a speaker, one or more LEDs, etc.).

(98) The control device **216** may perform one or more processes described herein. The control device **216** may perform these processes based on the processor **262** executing software instructions stored by a computer-readable medium, such as the memory **264** and/or the storage component **268**. A computer-readable medium (e.g., a non-transitory computer-readable medium) is defined herein as a non-transitory memory device. A memory device includes memory space located inside of a single physical storage device or memory space spread across multiple physical storage devices.

(99) Software instructions may be read into the memory **264** and/or the storage component **268** from another computer-readable medium or from another device via the communication interface **220**. When executed, software instructions stored in the memory **264** and/or the storage component **268** may cause the processor **262** to perform one or more processes described herein. Additionally or alternatively, hardwired circuitry may be used in place of or in combination with software instructions to perform one or more processes described herein. Thus, embodiments or aspects described herein are not limited to any specific combination of hardware circuitry and software.

(100) The memory **264** and/or the storage component **268** may include data storage or one or more data structures (e.g., a database, and/or the like). The control device **216** may be capable of receiving information from, storing information in, communicating information to, or searching information stored in the data storage or one or more data structures in the memory **264** and/or the storage component **268**. For example, the information may include input data, output data, or any combination thereof.

(101) The number and arrangement of components shown in FIG. 7 are provided as an example. In some non-limiting embodiments or aspects, the control device **216** may include additional components, fewer components, different components, or differently arranged components than those shown in FIG. 7. Additionally or alternatively, a set of components (e.g., one or more components) of the control device **216** may perform one or more functions described as being performed by another set of components of the control device **216**.

(102) With reference to FIGS. 8-9, a connector **200'** is shown in accordance with some non-limiting embodiments or aspects of the present disclosure. The components of the connector **200'** shown in FIGS. 8-9 are substantially similar or identical to the components of the connector **200** described

herein with reference to FIGS. 3-5. Reference numerals in FIGS. 8-9 are used to illustrate identical components of the corresponding reference numerals in FIGS. 3-5. As the previous discussion regarding the connector **200** generally shown in FIGS. 3-5 is applicable to the connector **200'** shown in FIGS. 8-9, only the relative differences between the two connectors are discussed hereinafter.

(103) With reference to FIG. 8, the frame **202** has a first end **202a** configured for connecting to a first piece of safety equipment, such as a safety harness **104** or a lanyard **110**, and a second end **202b** configured for connecting to a second piece of safety equipment, such as the lanyard **110** or anchor **106** (shown in FIG. 1). In some non-limiting embodiments or aspects, the connector **200'** is configured to be attached directly to the front/rear portion of safety harness **104** by way of a ventral/dorsal D-ring. The first and second ends **202a**, **202b** of the frame **202** may be configured as D-rings (FIG. 8) to allow connection to components of a fall protection assembly via a connection element **116**, such as a carabiner or a snap hook (not shown). In other embodiments or aspects, such as shown in FIG. 9, one of the first and second ends **202a**, **202b** may be a D-ring, while the other of the first and second ends **202a**, **202b** may be a slot configured for receiving, for example, a strap of the safety harness **104**, to allow the connector **200'** to be directly integrated with the safety harness **104**.

(104) With continued reference to FIGS. 8-9, the frame **202** of the connector **200'** may be made from metal. The frame **202** may be a forging or casting that has a central section **202c** for mounting the components of the connector **200'** that are covered by the housing **213** enclosing additional components of the connector **200'**. In some non-limiting embodiments or aspects, the housing **213** may be a cover formed from an epoxy or other fluid that hardens so that the components within the housing **213** cannot be damaged by normal usage and handling. In other embodiments or aspects, the housing **213** is a rigid cover that is removably or non-removably connected to the frame **202**. When the frame **202** is configured for use as a load-bearing component, the entire frame **202** may be configured to withstand a minimum force of 5000 lbs or 22 kN of force for repeated drop test.

(105) With continued reference to FIGS. 8-9, the sensor assembly **214** may be configured to detect whether the connector **200'** is connected to safety equipment, such as a component of a fall protection assembly. In some non-limiting embodiments or aspects, the one or more sensors **215** may be configured to detect whether the first end **202a** of the frame **202** of the connector **200'** is connected to the D-ring **108** of the harness **104** and whether the second end **202b** of the frame **202** is connected to the lanyard **110**. Connection between the connector **200'** and a component of the fall protection assembly, such as the harness **104** or the lanyard **110**, may be made by way of loops, D-rings, carabiners, locking snap hooks, or other connection mechanisms. With each of these connection mechanisms, the one or more sensors **215** of the connector **200'** is configured to detect whether a proper connection is made between these components and the first and second ends **202a**, **202b** of the frame **202**.

(106) In some non-limiting embodiments or aspects, the one or more sensors **215** is a Hall effect sensor configured for detecting whether the connector **200'** is connected to a piece of safety equipment. The Hall Effect sensor may be mounted flush with an outside surface of the housing **213**. The Hall Effect sensor may be surrounded by or positioned proximate to a magnetic surface. This magnetic surface may be configured to interact with a magnet **228** that is attached to a portion of a fall protection assembly, such as the safety harness **104** and the lanyard **110**. In some non-limiting embodiments or aspects, the separate magnets **228** may be permanently secured to the lanyard **110** and the harness **104** by way of a cable and grommet, or by way of a riveted connection.

(107) When the user connects the D-ring of the safety harness **104** to the lanyard **110** or other device, the connector **200'** is disposed therebetween such that the magnet **228**, either automatically or with additional guidance from the user, aligns and connects to the magnetic surface of the Hall effect sensor. This action changes the state of the Hall Effect sensor and indicates that the safety harness **104** is connected to the lanyard **110** via the connector **200'**. The length of the cable

connected to the magnet **228** is only as long as the distance between the Hall Effect sensor on the connector **200'** and its cabled anchor point on the lanyard **110**. The cable is sufficiently strong as to align with the connector **200'**, but flexible enough to bend as the connector **200'** is in use and the magnet is strong enough to stay attached under normal motion of the worker throughout the work day. The Hall Effect sensor may also be incorporated with the connection sensor assembly **214** described herein with reference to FIGS. 3-5.

(108) In some non-limiting embodiments or aspects, the one or more sensors **215** may be a strain gauge configured to detect connection of the safety harness **104** to the lanyard **110** or other safety equipment by way of pull strain at the connection interface of these components. One end of the strain gauge may be connected to a fixed anchor point or on the harness **104**, while the other end of the strain gauge may be connected to the lanyard **110**. In this manner, the connection of the lanyard **110** to the harness **104** is detected due to a strain reading on the strain gauge of the connector **200'**. The strain gauge may also be incorporated with the connection sensor assembly **214** described herein with reference to FIGS. 3-5.

(109) In some non-limiting embodiments or aspects, the one or more sensors **215** may be a contact switch configured to detect connection of the safety harness **104** to the lanyard **110** or other fall protection device. The contact switch may have a shield or arm that prevents a carabiner or other device to connect to the D-ring of the harness **104** or the connector **200'** without tripping the switch. The switch may be part of an assembly including an arm that partially covers the area within the D-ring. When the connector **200'** connects to the D-ring of the harness **104**, the arm or shield is moved away from its home position, thereby causing the switch to open or close depending on its configuration to indicate that a connection has been made between the D-ring of the harness **104** and the connector **200'**. A similar connection mechanism may exist between the lanyard **110** or other fall protection device and the connector **200'**. The contact switch may also be incorporated with the connection sensor assembly **214** described herein with reference to FIGS. 3-5.

(110) In some non-limiting embodiments or aspects, the one or more sensors **215** may be configured to detect movement of the connector **200'**, such as due to movement of the worker. In such embodiments or aspects, the one or more sensors **215** may be a gyroscope and/or an accelerometer. The gyroscope and/or accelerometer may also be incorporated with the connection sensor assembly **214** described herein with reference to FIGS. 3-5.

(111) In further embodiments or aspects, the one or more sensors **215** may include a gyroscope, an accelerometer, and a Hall Effect sensor. In this configuration, the gyroscope and accelerometer are configured for detecting a near zero acceleration event (fall), a high acceleration event (fall arrest), motion or lack of motion, and worker orientation. The accelerometer may be configured to detect motion even if the worker is in a relatively static position such as when the worker is seated, kneeling, or in a prostrate position. The Hall Effect sensor may detect whether the connector **200'** is connected to or disconnected from the fall protection assembly, as described herein. For example, in an event of a fall from an elevated surface, the unit connector **200'** may be configured to record the time and date of the event, along with the orientation and motion of the worker prior to, during, and after the fall event. In this way, specific alarms can be configured, broadcast, and recorded, regarding the state of the worker as being in distress or being able to recover from a possible fall event. If the accelerometers or gyroscopes show accelerations other than that of just gravity (e.g. the unit is NOT at rest, but is moving) then a second condition of compliant use of the connector **200'** is met. In any embodiment or aspect where the one or more sensors includes an accelerometer or a gyroscope, data from these sensors can be used to show a pattern of use or behavior regarding the device over time. In embodiments or aspects where the connector **200'** has a gyroscope, the initial orientation of the gyroscope can be determined and saved as a baseline orientation. Deviation in orientation from this baseline orientation can be used for generating an alert that the user may have fallen or is in a situation where assistance may be required. The combined

gyroscope, an accelerometer, and a Hall Effect sensor may also be incorporated with the connection sensor assembly **214** described herein with reference to FIGS. **3-5**.

(112) In some non-limiting embodiments or aspects, the loop-shaped first and second ends **202a**, **202b** of the connector **200'** may have a retractable cover panel **230** movable between a retracted position and an extended position. In the retracted position, the loop-shaped ends **202a**, **202b** are open to allow connection of the connector **200'** to another device, such as the D-ring of the harness **104** or the lanyard **110**. In the extended position, the loop-shaped ends **202a**, **202b** are covered to prevent connection of such devices to the connector **200'**. The retractable panel **230** may be configured to be automatically moved from the extended position to the retracted position when a D-ring, carabiner or other connector used in fall protection applications is attached to either end **202a**, **202b**.

(113) The retractable panel **230** may be connected to a sensor, such as a mechanical, electrical, or an optical sensor, that is configured to detect the position of the cover panel **230**, such as the retracted position, the extended position, or an intermediate position between the extended position and the retracted position. In some non-limiting embodiments or aspects, connection of the D-ring, carabiner or other connector used in fall protection applications to the connector **200'** may move the panel **230** from the extended to the fully retracted position in a direction of arrow B in FIG. **8**, thereby indicating a proper connection. The panel **230** may be biased to the extended position by a biasing mechanism, such as a spring. If the panel **230** is moved to the intermediate position, this may indicate that an improper connection is made or that an improper connector is connected to the connector **200'**. The connector **200'** may be configured to detect whether a connection has been made to one or both of the ends **202a**, **202b** of the frame **202** based on a position of the panel **230**. If the connector **200'** detects that both ends **202a**, **202b** are connected to a respective connector, this information may indicate a first condition of compliant use of the connector **200'**.

(114) In some non-limiting embodiments or aspects, a deflectable element **232**, such as a paddle, may be disposed within an open area of the first and second ends **202a**, **202b** of the frame **202** of the connector **200'**. The deflectable element **232** may be configured to pivot around the axis that is perpendicular to the housing **213** between an undeflected, or closed position and a deflected, or open position. The deflectable element **232** may be biased to the closed position by a biasing mechanism, such as a spring. The deflectable element **232** may interact directly or indirectly with a switch, such as a contact, optical, magnetic or other type of switch. In some non-limiting embodiments or aspects, the switch may be a non-contact type such that the housing **213** can be completely sealed from the surrounding environment. When a connection is made to the first or second end **202a** **202b** of the frame **202**, the deflectable element **232** is moved from its initial position to a second, deflected position. The connector **200'** may be configured to detect whether a connection has been made to one or both of the ends **202a**, **202b** of the frame **202** based on a position of the deflectable element **232**. If the connector **200'** detects that both ends **202a**, **202b** are connected to a respective connector, this information may indicate a first condition of compliant use of the fall protection assembly.

(115) The connector **200'** may be configured to store data received from one or more sensors **215**, along with time, date, and/or location data associated with data received from one or more sensors **215**. The connector **200'** may have an acquisition mode wherein data received by one or more sensors **215** is stored in system memory along with a time, date, and/or location stamp. When in the acquisition mode, the connector **200'** may store any conditional change of the one or more sensors **215**, along with a time, date, and/or location stamp. For example, turning the connector **200'** on and placing it in the acquisition mode may save in the system memory an initial time, date, and/or location stamp. Connection of one end **202a**, **202b** of the frame **202** to a connector generates a conditional change in the one or more sensors **215**, which is saved in the system memory along with the time, date, and/or location stamp associated with such conditional change. Similarly, if the connector **200'** is in motion due to movement of the worker, such conditional change may be saved

in the system memory along with the time, date, and/or location stamp associated with such conditional change. Data recorded to system memory can be organized in any manner of logs or intervals from minutes, hours, shifts, days, weeks, or similar.

(116) Referring now to FIG. 10, FIG. 10 is a diagram of an example environment 400 in which the fall protection compliance system 100 described herein may be implemented. As shown in FIG. 10, environment 400 includes the fall protection compliance system 100, which may include a plurality of connectors 200, various pieces of safety equipment, such as lanyards, to which the connectors 200 may be connected, a plurality of anchors having one or more connection structures 300 with identification elements 320. In some non-limiting embodiments or aspects, the environment 400 may correspond to a construction site where a plurality of users may be assigned to work. The environment 400 may further include a communication network 402, and one or more remote devices 250 configured to communicate via one-way or two-way communication with the connectors 200 of the fall protection compliance system 100 using the communication network 402. In some non-limiting embodiments or aspects, the plurality of connectors 200 may be configured to communicate with each other using the communication network 402. The remote device(s) 250 and the connectors 200 may interconnect (e.g., establish a connection to communicate, and/or the like) via wired connections, wireless connections, or a combination of wired and wireless connections using the communication network 402.

(117) In some non-limiting embodiments or aspects, the environment 400 allows authorized users to perform preventive occupational health and safety actions and manage inspections and maintenance of safety equipment. Each remote device 250 may be configured to transmit and/or receive data to and/or from the connectors 200 of the fall protection compliance system 100 via a short-range wireless communication connection (e.g., an NFC communication connection, an RFID communication connection, a Bluetooth® communication connection, and/or the like). In some non-limiting embodiments or aspects, the remote device 250 may be associated with a user (e.g., a safety manager).

(118) Communication network 402 may include one or more wired and/or wireless networks. For example, the communication network 402 may include a cellular network (e.g., a long-term evolution (LTE) network, a third generation (3G) network, a fourth generation (4G) network, a fifth generation (5G) network, a code division multiple access (CDMA) network, etc.), a public land mobile network (PLMN), a local area network (LAN), a wide area network (WAN), a metropolitan area network (MAN), a telephone network (e.g., the public switched telephone network (PSTN)), a short-range wireless communication network (e.g., an NFC communication network, an RFID communication network, a Bluetooth® communication network, and/or the like), a private network, an ad hoc network, an intranet, the Internet, a fiber optic-based network, a cloud computing network, and/or the like, and/or a combination of some or all of these or other types of networks.

(119) The number and arrangement of systems and/or devices shown in FIG. 10 are provided as an example. There may be additional systems and/or devices, fewer systems and/or devices, different systems and/or devices, or differently arranged systems and/or devices than those shown in FIG. 10. Furthermore, two or more systems and/or devices shown in FIG. 10 may be implemented within a single system or a single device, or a single system or a single device shown in FIG. 10 may be implemented as multiple, distributed systems or devices. Additionally, or alternatively, a set of systems or a set of devices (e.g., one or more systems, one or more devices) of the environment 400 may perform one or more functions described as being performed by another set of systems or another set of devices of the environment 400.

(120) With reference to FIGS. 11-12, data received by the remote device 250 may be displayed on a graphical user display 600 for review by the compliance manager, industrial hygienist or other supervisor personnel. The received data may include, for example, the time-stamped transitions between all device states, sensor data, and device health status. For example, as shown in FIG. 11,

the display **600** may include information regarding the connection status of each connector that is part of the fall protection compliance system **100**. In some non-limiting embodiments or aspects, such as shown in FIG. **12**, the display **600** may include information regarding the user's name or the connector name, battery life, time of current connection, current date and time, total time tied off (accumulated since the connector was turned on), total number of times the connector has been tied off, total time (accumulated since the device was turned on), location of specific connecting structure that the connector is connected to (dictated by the RFID tag), altitude (dictated by an altimeter on the device board), and any combination thereof.

(121) Referring now to FIG. **13**, FIG. **13** is a flowchart of a non-limiting embodiment or aspect of a process **500** for determining a connection status of a connector **200**. In some non-limiting embodiments or aspects, one or more of the functions described with respect to process **500** may be performed (e.g., completely, partially, etc.) by the connector **200**. In some non-limiting embodiments or aspects, one or more of the steps of process **500** may be performed (e.g., completely, partially, and/or the like) by another device or a group of devices separate from the connector **200**, such as the remote device **250**.

(122) With continued reference to FIG. **13**, at step **502**, the process **500** may include receiving, with the control device **216**, data gathered by one or more sensors **215**. For example, in some non-limiting embodiments or aspects, the process **500** may include receiving connection data gathered by the at least one magnetometer **217**. Detecting the presence of the disturbance in the magnetic field may be indicative of a presence of a ferromagnetic material of the connection structure **300** within the connection area **206** of the connector **200**. Conversely, detecting the absence of the disturbance in the magnetic field may be indicative of an absence of the ferromagnetic material of the connection structure **300** within the connection area **206** of the connector **200**. As described herein, the at least one magnetometer **217** is configured to detect a presence or an absence of a disturbance in a pre-defined magnetic field within the connection area **206** generated by the permanent magnet. The connection data may include, for example, a signal that corresponds to the characteristics of a magnetic field detected by the at least one magnetometer **217**.

(123) With continued reference to FIG. **13**, at step **502**, the process **500** may further include receiving, with the control device **216**, communication data gathered by the at least one short-range wireless communication antenna **221**. Detecting the presence of the identification element **320** may be indicative of a presence of an approved connection structure **300**. Conversely, detecting the absence of the identification element **320** (i.e., not detecting the identification element **320**) may be indicative of an absence of the approved connection structure **300**. As described herein, the at least one short-range wireless communication antenna **221** is configured to detect a presence or an absence of an identification element **320** that is connected to the connection structure **300**. The communication data may include, for example, a signal that is indicative of whether the identification element **320** is within the detection range of the at least one short-range wireless communication antenna **221**.

(124) With continued reference to FIG. **13**, at step **504**, the process **500** may include determining, with the control device **216**, a connection status of the connector **200** based on detecting the presence or the absence of the disturbance in the magnetic field via the at least one magnetometer **217**. For example, by detecting the presence of the disturbance in the magnetic field via the at least one magnetometer **217**, the control device **216** may correlate such disturbance to be indicative of a presence of a ferromagnetic material of the connection structure **300** within the connection area **206** of the connector **200**. Conversely, by detecting the absence of the disturbance in the magnetic field via the at least one magnetometer **217**, the control device **216** may correlate such absence of the disturbance to an absence of the ferromagnetic material of the connection structure **300** within the connection area **206** of the connector **200**.

(125) With continued reference to FIG. **13**, at step **504**, the process **500** may further include determining, with the control device **216**, a connection status of the connector **200** based on

detecting the presence or the absence of the identification element **320** on the connection structure **300** via the at least one short-range wireless communication antenna **221**. For example, by detecting the presence of the identification element **320** via the at least one short-range wireless communication antenna **221**, the control device **216** may correlate such presence to be indicative of a presence of an approved connection structure **300**. Conversely, by detecting the absence of the identification element **320** (i.e., not detecting the identification element **320**) via the at least one short-range wireless communication antenna **221**, the control device **216** may correlate such absence to be indicative of an absence of the approved connection structure **300**.

(126) With continued reference to FIG. **13**, at step **506**, the process **500** may include performing, with the control device **216**, at least one action based on the connection status. For example, the control device **216** may provide a first visual, audio, and/or tactile indication that is indicative of a presence of a ferromagnetic material of the connection structure **300** within the connection area **206** and/or a presence of an identification element **320** on the connection structure **300**. Conversely, the control device **216** may provide a second visual, audio, and/or tactile indication that is indicative of an absence of the ferromagnetic material of the connection structure **300** within the connection area **206** and/or an absence of an identification element **320** on the connection structure **300**. In further examples, the control device **216** may be configured to transmit the connection status to a remote device **250** via the communication network **402**.

(127) The language used in the specification has been principally selected for readability and instructional purposes, and it may not have been selected to delineate or circumscribe the inventive subject matter. It is therefore intended that the scope of the disclosure be limited not by this detailed description, but rather by any claims that issue on an application based hereon. Accordingly, the disclosure of the embodiments of the disclosure is intended to be illustrative, but not limiting, of the scope of the disclosure, which is set forth in the following claims.

(128) As noted above, a connection structure as described herein can take a variety of forms. FIG. **14** is a diagram depicting a hook connection structure **1400** having a plurality of sensors provided thereon. The connection structure includes a hook portion **1402**, the hook portion **1402** being connected to a connection point **1404** for connecting the connection structure **1400** to an external structure, such as a line that is connected to a rigid structure, a line that is connected to a user (e.g., a harness the user is wearing), or directly to a user, such as via a harness. The hook portion **1402** includes a base, hook shaped structure that includes an end portion **1406** for receiving an anchor structure (not shown in FIG. **14**). As described above, that anchor structure may be directly connected to a rigid structure (e.g., via a hook, a ring, a carabiner, a line loop), indirectly to a rigid structure via an intervening structure (e.g., a line, an energy absorber, additional hooks or other connectors), or to a user directly or indirectly. The hook portion **1402** further includes a movable gate **1408** for completing a closed loop that prevents an anchor structure captured therein from disengaging from the hook portion **1402**. As described above, the movable gate **1408** may be biased toward the closed position as depicted in FIG. **14** or otherwise secured (e.g., via a gate lock **1410**, via a threaded bolt) such that the possibility of accidental breaking of the closed loop provided by the hook portion **1402** is mitigated.

(129) The hook connection structure **1400** of FIG. **14** includes certain electronics and sensors for monitoring a state of connection of the connection structure **1400**. The hook portion **1402** is equipped with electronics **1412** for receiving sensor signals from a set of sensors **1414**, **1416**, **1418**, **1420**, storing data associated with those sensor signals for later access, and for transmitting that stored data to an external entity, such as via a wired or wireless connection. In embodiments, the electronics **1412** may further include a battery (not shown) for powering the electronics **1412** and sensors and functionality for providing status and alarm information (e.g., lights, a speaker). In the example of FIG. **14**, the hook portion **1402** includes an interlock switch **1414** for monitoring the status of the movable gate **1408**. In embodiments, detection of movement of the gate **1408** via switch **1414** can be used as a wakeup signal instructing the electronics **1412** to begin actively

collecting data from the attached sensors. In embodiments, a lack of sensor activity for more than a timeout period of time (e.g., minutes, a half hour, an hour, hours) may be interpreted as indicating that the connection structure **1400** is not in use and as a command to put the connection structure **1400** into a low power or sleep state to prolong battery life. Data regarding the position of the movable gate **1408** acquired via the switch **1414** may be monitored and stored by the electronics **1412** (e.g., open status and associated time stamp, closed status and associated time stamp). An alarm condition may be noted and acted upon (e.g., with an audible alarm) when the hook structure **1402** is in an abnormal state, such as when the switch **1414** indicates that the movable gate **1408** is in the open position for a prolonged period of time (e.g., more than 10 seconds, more than a minute), which may indicate that a user is not properly connected (e.g., a line or other anchor structure is pinned between the movable gate **1408** and the hook portion **1402**).

(130) The hook structure **1402** may further be equipped with one or more accelerometers **1416**, **1418**. In the example of FIG. **14**, a first accelerometer **1416** provides data regarding three dimensional relative motion of the connection structure **1400**. The first accelerometer **1416** provides data regarding movement in one or more of an x, y, and z direction. The accelerometer **1416** data can be stored (e.g., via periodic, timestamped sampling on a microsecond, millisecond, second, minute basis) and transmitted from the hook portion **1402**. That data may be interpreted by the electronics **1412** or remotely interpreted for indications of events such as a fall event and a fall arrest event. Alarms may be issued upon detection of such events. Data from the accelerometer **1416** may further be used to detect compliant continued connection of the connection structure to an anchor structure, alone or in combination with data from the other sensors. For example, changes to the accelerometer data may indicate that a user is currently properly connected, as opposed to a user who has clipped the connection structure **1400** to an anchor structure, where the connection structure **1400** is not actually connected to a user's harness (e.g., the connection structure **1400** is in a dead hang configuration). Further, a signature of data from the accelerometer **1416** may be used to indicate anomalous conditions. For example, accelerometer data that is periodic in nature could be interpreted to indicate an improper connection (e.g., a dead hanging connection structure swinging in a windy environment). An alarm may be issued if the electronics **1412** identify an improper connection. Further, such an improper connection could be later identified (e.g., after wireless transmission of data), such that corrective action can be taken (e.g., further training of the user, reprimand).

(131) The example of FIG. **14** further includes a second accelerometer **1418** in the form of a piezo sensor. The example piezo sensor **1418** provides a charge and associated voltage when mechanical stress is applied to its structure. In the example of FIG. **14**, the piezo sensor **1418** comprises an insulated wire that is positioned along an inside edge of the end portion **1406** where the piezo sensor **1408** and/or surrounding structure (as described further below) mechanically interacts with an anchor structure positioned within the end portion **1406** of the hook portion **1402**. A voltage signal generated by the piezo sensor **1406** accelerometer is received by the electronics **1412** for storage as data, interpreted for detection of events, or transmission to an external entity such as a remote server. In embodiments, piezo sensor **1418** data can be used to detect events, alone or in combination with other sensor data. For example, a lack of a signal from the piezo sensor **1418** could indicate that a user is not currently connected properly to an anchor structure because of the lack of relative movement and contact of the connection structure **1400** relative to the anchor structure. Data from the piezo sensor **1418** can further be interpreted to indicate active events such as a fall, a fall arrest, or movement along a line. For example, the electronics **1412** or an external reviewing processor can be trained to recognize signatures in the piezo data indicative of event data. For example, a sudden halt of a piezo signal may indicate a free fall condition. A sudden spike could indicate a fall arrest condition. A periodic signal could indicate a user movement event (e.g., indicating the connection structure **1400** scraping across a horizontal line anchor structure). Alarms may be issued or future corrective action taken based on the detection of events locally at the

electronics **1412** or remotely after data transmission.

(132) The hook portion **1402** of FIG. **14** further includes a pressure switch **1420** that collects data regarding the existence and/or amount of pressure experienced at the sensor, such as through a force applied by the anchor structure interacting with the hook portion **1402**. For example, the pressure switch **1420** may be positioned next to a spring surrounding the piezo sensor **1418**, as described further below. Metallic contact between the spring and the pressure switch (e.g., when the anchor structure presses the spring against the end portion **1406**) may provide a conductive path enabling transmission of a signal to the electronics **1412**. The pressure switch may be used alone or in combination with other sensor data to indicate events. For example, an active pressure switch signal may indicate that the anchor structure is actively pressing against the end portion **1406** of the connection structure **1400**, indicating that a user is currently supported by the connection structure **1400**. This may be indicative of a proper connection on its own. In other examples, in combination with data indicating sudden movement (e.g., from accelerometers **1416**, **1418**), an active pressure switch **1420** signal could be indicative of a fall arrest condition, where appropriate alarms or other corrective action can then be taken.

(133) As noted above, each of the sensors **1414**, **1416**, **1418**, **1420** provide data that provides some indication on whether a user utilizing the connection structure **1400** is continually connected to the anchor structure. In embodiments, the electronics **1412** on the connection structure **1400** are configured to determine based on data from one or more of the sensors whether a user is currently properly connected to the anchor structure. The electronics **1412** are configured to store data associated with that sensed continued connection of the connection structure **1400** to the anchor structure (e.g., a Boolean yes/no connection signal, a likelihood of proper connection value, a connection type value (e.g., no connection, connection with active movement, connection with limited movement)). That data can be output from the connection structure to an external entity for monitoring compliance, and in embodiments can be used to issue an alarm (e.g., an audible alarm when a lack of connectivity is detected to prompt a user to reconnect).

(134) FIG. **15** is a diagram depicting an example configuration of an interlock switch configured to monitor a position of a movable gate of a hook connection structure. As described above, a hook connection structure includes a moveable gate **1408** that enables interfacing of the connection structure with an anchor structure. In embodiments, the movable gate **1408** may be held in place by a gate lock **1410** to prevent unintentional movement of the movable gate **1408**. An interlock switch **1414** is configured to monitor the position of the movable gate **1408** to enable capture of data regarding whether and at what times the movable gate **1408** is in the open and closed position, such as in an event log data structure.

(135) As noted above, a fall protection compliance system may monitor continued connection of the connection structure to the anchor structure over a period of time and to output a signal indicating that continued connection, such as logic embedded in electronics associated with the connection structure. In one embodiment, the logic dictates that a connection or disconnection event will be preceded by a signal from the gate switch **1414**. A user depresses gate lock **1410** while simultaneously moving the movable gate **1408** to transition the movable gate **1408** to the open position. This allows the user to connect or disconnect the connection structure from the anchor structure. When the movable gate **1408** is moved to the open position, an edge **1502** of the movable gate structure contacts a lever on the interlock switch **1414**, actuating the switch. As noted above, a signal from the switch **1414** indicating a move to the open position can act as a wakeup signal for the electronics of the connection structure, such that the electronics begin to actively seek to detect an initial and ongoing connection of the connection structure with the anchor structure. An open gate signal, followed by a closed gate signal, in combination with signals from the other sensors (e.g., an accelerometer, a piezo sensor, a pressure switch) can be deemed an indicator that a connection has been established. Further signals from those other sensors can indicate continued connection. A subsequent open gate signal from the gate switch **1414** could be indicative of a

disconnect event. For example, a subsequent open gate switch followed by limited or no further signals from other sensors could be interpreted by the electronics as a disconnect event for storage in a memory, log file, or transmission to an external entity.

(136) FIG. **16** is a diagram depicting an example configuration of a piezo sensor and a pressure switch. FIG. **16** depicts a hook connection structure having an end portion **1406** that is interfaced with an anchor structure **1602**, where a cross section of that anchor structure (e.g., a ring, a carabiner) is depicted in FIG. **16**. The example of FIG. **16** includes a piezo sensor **1418** and a pressure switch **1420**. The piezo sensor **1418** takes the form of an insulated piezo cable positioned within a protective metallic spring **1604**. A voltage is induced in the piezo cable **1418** when that cable is deformed, experiences movement or experiences shock, such as by contact of the anchor structure **1602**. As described further herein, the piezo sensor **1418** may be positioned within a deformable portion of the end portion **1406** such that relative movement of the piezo sensor **1418** and surrounding protective spring **1604** is possible in relation to the anchor structure **1602**. The induced voltage in the piezo sensor is transmitted via a wire **1606** to the electronics of the connection structure, for sensing of a status of the connection structure. FIG. **16** further includes a pressure switch **1420**. In the example of FIG. **16**, the pressure switch is formed from the combination of the metallic contact plate **1610** and the protective metallic spring **1604**. The bend in the spring induces compression on the spring side towards **1602** and tension on the opposite side towards **1420**. The compression/tension is such that a Force F is required to stretch and move the spring towards the metallic contact plate **1420**. In operation, presence of the anchor structure **1602** in the end portion **1406** can impart a force on the protective metallic spring **1604** toward the end portion **1406**. That force can move the protective metallic spring toward the metallic contact plate **1610** such that metallic contact is made between the two. When metallic contact is made, an electrically conductive circuit is formed which transits over the wire pair **1606** to the electronics **1412** which detects the conductive path formed between the wire pair **1606** thus sensing a connection status of the connection structure.

(137) FIG. **17** is a diagram depicting an example configuration of a piezo sensor and a pressure switch. The piezo sensor, depicted at **1700**, comprises a piezo coaxial type of cable **1702**. The piezo cable **1702** is connect to electronics **1704** of the connection structure, such that a voltage induced in the piezo cable **1702** via deformation, movement, or shock is transmitted to the electronics **1704** for use in detecting connection status of the connection structure. The piezo cable **1702** is positioned within the protective spring **1706**. The piezo cable **1702** and surrounding protective spring **1706** (when assembled) are positioned between an inner deformable structure **1708** and an outer deformable structure **1710**. In embodiments, the deformable structures **1708**, **1710** are made from non-conductive material, such as ABS or nylon plastic such that they provide structure for the conductive wire **1702** and protective spring **1704** to sit within while not interfering with communication of data signals.

(138) A pressure switch is depicted at **1750**. There, the protective spring **1706** having the piezo sensor piezo cable therein (connection of the piezo cable to the electronics **1704** not shown) is set for positioning between the inner **1708** and outer **1710** deformable structures. The pressure switch depiction at **1750** illustrates the protective spring **1706** being connected to a first wire **1752** that connects to the electronics of the connection structure. A contact plate **1754** is positioned within the outer deformable structure **1710**, with the contact plate **1754** being connected to a second wire **1756** that connects to the electronics of the connection structure. In operation, when an anchor structure is within the end portion of the connection structure, the anchor structure can impart a force on the inner deformable structure **1708**. That force pushes the protective spring **1706** such that it makes contact with the contact plate **1754**. This completes a circuit through wire **1752**, protective spring **1706**, contact plate **1754**, and wire **1756** to the electronics, providing a signal indicating pressure being applied at the pressure switch. In another embodiment the positions of the contact plate and protective spring relative to the anchor structure may be reversed such that as the

anchor structure imparts a force on the inner deformable structure, and the contact plate makes contact with the protective spring. In examples, the shape of the deformable structures may be augmented to provide desired characteristics of sensor output signals. For example, a fulcrum structure **1758** can be implemented on the inner deformable structure **1718** such that an output of the piezo sensor **1700** and/or pressure sensor **1750** provides a recognizable signature under certain operational scenarios.

(139) As noted above, signals from the various sensors that may be integrated with the connection structure can be used to identify connection of the connection structure to an anchor structure. In addition to a Boolean connected/not connected detection, a connection structure or external processor analyzing signals therefrom may detect more refined indications of connection status and events. For examples, signatures of different known connection statuses (e.g., not connected, a stationary (sitting, standing, kneeling) connected user, a moving connected user, a connection structure not connected to a user) can be compared to current output from one or more sensors to provide an indication of a type of current connection. FIG. **18** is a diagram depicting example piezo sensor outputs when a connection structure is provided in different operating conditions. Training of an analyzing processor (e.g., an external processor, a processor embedded in the connection structure electronics), such as via machine learning, can enable detection of operational states of the connection structure. For example, repeated training of a system using tagged data indicative of a fall condition can enable a processor to detect a fall condition during live use. Similarly, a fall arrest, connected stationary user, a connected moving user, and anomaly conditions can be trained and detected.

(140) In the example of FIG. **18**, a first signature **1802** from a piezo sensor without a fulcrum indicating a fall condition is depicted. A second signature **1804** is from a piezo sensor having a fulcrum (see FIG. **17** at **1758**) indicating a fall condition is depicted (e.g., a force greater than 1000 pounds). A third signature **1806** indicates a connected, stationary user. A fourth signature **1808** indicates a connected user experiencing a large bump condition (e.g., a force between 500 pounds and 1000 pounds). A fifth signature **1810** indicates a connected user experiencing a small bump condition (e.g., a force between 100 pounds and 500 pounds). A sixth signature **1812** indicates a disconnection condition, where a user disconnects an anchor structure from the connection structure.

(141) FIG. **19** is a graph depicting a signature associated with a connected user moving relative to a horizontal support line anchor structure. At a left edge of the graph, an initial pulse **1901(a)** is illustrated generated by initial contact of the anchor structure with the protective spring. Following are four pulses **1901(b)**-**1901(e)** that are generated as the anchor structure drags across the pitch of the spring as the connection structure hook moves along the horizontal wire. In embodiments, the connected, moving user signature can be validated via use of data from another sensor, such as another accelerometer that indicates motion of the connection structure or a gate switch that indicates that a connection of the connection structure to the anchor structure may have been made such that the detected signature of FIG. **19** is valid.

(142) In addition to recognition of particular waveforms as indicating connectivity status, the continued presence of an output signal from a piezo sensor can be utilized. FIG. **20** is a diagram depicting an example piezo sensor output waveform **2001** indicative of a connected user. In one example, continued connection of a user is detected using a leaky bucket algorithm. In that algorithm, the exact shape of the waveform is less important than the presence (or lack) of a varying signal from the piezo sensor over time. In the leaky bucket algorithm, the signal is integrated over a period of time (e.g., the period of time depicted in FIG. **20**). That integrated value is compared to a threshold, wherein a value over that threshold may be deemed indicative of a level of motion attributable to a connected user. As time passes, a portion of the waveform moves out of the integration window, such that it is no longer considered in the comparison to the threshold. Thus, once a level of motion over a period drops below a certain level on average, the threshold is

not met and the system may generate an alert or other indication that a user may no longer be connected to the anchor structure.

(143) A process for providing fall protection compliance monitoring may take a variety of forms. FIG. 21 is a flow diagram depicting an example method of providing fall protection compliance monitoring. A system begins at **2102** in a sleep state. When a signal is received from the interlock switch, indicating movement of the movable gate of the connection structure, circuitry including other sensors is transitioned to a wake state at **2104**. Those sensors are used to acquire data at **2106**. Data from those sensors is monitored for a period of time via a loop from **2108** to **2106** until a connection is detected (e.g., based on signals from an accelerometer, piezo sensor, or pressure sensor indicating interaction with an anchor structure) where the system progresses to **2110** or until a timeout period occurs where the system then returns to the sleep state at **2102**. When a connection is detected, that connection may be recorded in memory (e.g., with a date/time stamp) and/or reported externally, such as via a wireless data transmission. At **2112**, continued connection verification is performed, such as via monitoring of connected sensors (e.g., sensors other than the gate switch). A Boolean connected signal or more detailed connection type characterization may be recorded in memory with a date/time stamp and may be transmitted wirelessly via the loop at **2110**, **2112**. At **2114**, a subsequent activation of the gate interlock switch may be detected. This could be an indication that the connection structure is being disconnected from an anchor structure. At **2116**, data (e.g., data from other sensors of the connection structure) is monitored to see if that data is indicative of connection termination (e.g., no or minimal acceleration data, no pressure sensor data, no or minimal piezo sensor data). If the sensors indicate a disconnect, then data regarding that disconnect (e.g., a date/time stamp) is recorded in memory and/or reported externally, such as via a wireless data transmission, while the device returns to a sleep state at **2102**. If further sensor data indicates that a connection remains, then the interlock switch activation may be recorded and/or reported, with the process returning to **2112** for further monitoring.

(144) FIG. 22 is a flow diagram depicting another example method for providing fall protection compliance monitoring. At **2202**, a signal from a connection sensor associated with a connection structure configured for interfacing with an anchor structure is received, where the connection sensor comprises an accelerometer or a pressure switch. At **2204**, a determination is made as to whether a user is continually connected over a period of time based on the signal, and at **2206** a signal is output indicating said determination of whether the user is continually connected.

(145) Alternate examples of components and concepts herein are also within the scope of this disclosure. For example, mechanisms for monitoring the status of a gate structure of a hook may take a variety of forms. FIGS. 23A-H depict examples of gate structure monitoring sensors that provide signals indicating the status of the gate structure. FIG. 23A depicts a limit switch sensor. In FIG. 23A, a limit switch **2302** assumes one position when the movable gate **1408** is in a closed position and a second position when the gate **1408** is moved to an open position. For example, rotation of the movable gate **1408** may compress the limit switch **2302** when the gate **1408** is in the open position, while the limit switch **2302** returns to an uncompressed position when the gate **1408** is again closed. FIG. 23B depicts a pivot arm and a reed switch arrangement. There a torsion spring **2304** is positioned next to the movable gate **1408** such that it is rotated in the clockwise direction when the gate **1408** is moved to the open position. When rotated in that clockwise direction a magnet **2306** positioned on the torsion spring is moved closer to a reed switch **2308**, which detects the presence of the magnet **2306**. The reed switch **2308** may provide a binary signal indicating the gate **1408** is open or closed, or in another embodiments, another type of sensor, such as a hall effect sensor, may provide a more detailed data value that can be used to infer an amount of opening (e.g., 10%, 50%, 100%) of the movable gate **1408**.

(146) FIG. 23C depicts an alternate positioning of a magnet for interaction with a reed switch sensor. In this example, the magnet **2306** is affixed directly to the movable gate **1408**, such as via adhesive, in the top diagram, and as a component of a bracket **2310** in the bottom diagram. When

the movable gate **1408** is moved from its closed position, a reed switch (not shown) on the body of the hook detects the nearer-presence of the magnet **2306** and records and/or transmits a signal in response. FIG. **23D** illustrates a further positioning of a magnet **2306** on a portion **2312** of hook that is rotated clockwise (e.g., a gate lock), and closer to a reed switch **2308** when the movable gate **1408** is moved from a closed to an open position.

(147) FIG. **23E** provides a further example position of a limit switch. In FIG. **23E**, the limit switch **2302** is positioned adjacent to the rotatable portion **2312** of the hook illustrated in FIG. **23D**.

Rotation of that portion **2312** of the hook structure compresses the limit switch **2302** generating a signal that indicates that the movable gate **1408** is currently open.

(148) FIG. **23F** illustrates a further reed switch embodiment, where the magnet **2306** is placed on the rotatable portion **2312** of the hook. A reed switch **2308** is positioned near the top of that portion **2312**. When the movable gate is moved to the open position, the rotatable portion **2312** rotates clockwise, changing the pole direction of the magnet **2306** relative to the reed switch **2308**. This changes the magnetic force detected by the reed switch **2308**, where the reed switch generates a signal indicating a position of the gate based on that detected change in magnetic force.

(149) FIG. **23G** depicts a mechanical double throw switch **2314** positioned near the tip of the hook structure. In this example, an arm of the switch **2314** is pushed in a first direction (right) when the movable gate **1408** transitions from the closed position to the open position. The arm of the switch **2314** is pushed in a second direction (left) when the movable gate **1408** is transitioned back to the closed position. Signals indicating a current position of the movable gate **1408** are generated based on the arm of the switch **2314** being transitioned to either the right or left state by the movable arm **1408**.

(150) FIG. **23H** depicts a further reed switch embodiment, where a magnet **2306** is positioned on a torsion arm **2316** that is attached to a body **2320** of the hook. When the rotatable portion **2312** of the hook rotates counter-clockwise on opening of the hook, the reed switch **2308** moves relative to the magnet **2306**, generating a signal indicative of the gate opening.

(151) In another embodiment, the torsion arm **2316** may be connected to the rotatable portion **2312** of the hook that rotates clockwise when the movable gate **1408** is moved to the open position. That gate opening motion brings the magnet **2306** near to the reed switch **2308**, which generates a signal indicating the current position of the movable gate **1408**. In an embodiment, the torsion arm is biased by a spring, such that physical interaction of the magnet **2306** with the reed switch **2308** compresses the spring, generating a counterclockwise force, pushing the structure to close the movable gate **1408**.

(152) In some embodiments, it may be beneficial to have a housing within the connection structure that is able to rotate with the connector. For example, such an arrangement may keep close proximity between an identification element positioned within the housing and a connection sensor assembly of a connector. That close proximity can increase the reliability of reads of the identification element by the connection sensor assembly when a connector is properly connected to a connection structure having a housing therein.

(153) FIG. **24** depicts an example connection structure **2400** having a connection loop **2402** that is configured for being received within the connection area **206** of the connector **200** when the movable gate **204** is in the open position. The connection structure **2400** further has an anchor point **2404** configured as a separate loop for connecting to an anchor or a harness. The connection structure **2400** has a housing **2406**, which may be fixed within the connection loop **2402** such that there is an aperture **2408** formed by the connection loop **2402** and the housing **2406** (shown in further detail in FIG. **25**) configured to receive the connector **200** as described above. In embodiments, the housing **2406** is configured to hold an identification element **2420** such that the identification element is held close to a connection sensor assembly **214** when a connector is properly connected, allowing the connection sensor assembly to reliably sense the identification element (e.g., an RFID reader in the connection sensor assembly reading an identification element

2420 in the form of an RFID tag). A close up view of an example connection structure **2400** is shown in FIG. **25**, where the connection structure **2400** is implemented as a D-ring connector to an anchor or a harness.

(154) With further reference to FIG. **24**, a connector **200** may be configured to detect whether a connection is made with the connection structure **2400**, such as by detecting a presence of an identification element (e.g., a passive RFID tag) positioned within a housing **2406** in the connection structure **2400**. The connection structure **2400** may interact with the connector **200** in a similar manner as connection structure **300** and the elements of connection structure **2400** may operate in a similar manner as the elements of connection structure **300**. Specifically, in one example, the aperture **2408** formed by the connection structure **2400**/housing **2406** are configured to receive a connector **200** therein when the connector **200** is connected to the connection structure **2400**. When the connector **200** is within the aperture **2408**, the connection sensor assembly **214** is positioned in close proximity to the identification element **2420** (e.g., a passive RFID tag), permitting reliable reading of the identification element **2420** by the connection sensor assembly **214**. A successful reading of the identification element **2420** by the connection sensor assembly **214** can be used as an indication that the connector **200** has been properly connected to the connection structure **2400** associated with the identification element **2420**.

(155) In embodiments, the housing **2406** is able to rotate within the connection loop **2402** as pushed by the connector **200** connected within the aperture **2408**, maintaining close proximity between the identification element **2420** and the connection sensor assembly **214** for reading the identification element **2420** while allowing freedom of movement for the operator connected to the connection structure **2400**.

(156) A housing, as depicted in FIGS. **24** and **25** may take a variety of forms. For example, FIG. **26** depicts a housing **2406** that includes a first component **2510** and a second component **2520**. In the example of FIG. **26**, the first component **2510** and second component **2520** are configured to be snapped together when connected through the connection loop **2402** such that the two snapped-together components **2510**, **2520** grip the connection loop **2402** there between via a groove formed by the connected components **2510**, **2520**. In embodiments, the first component **2510** and the second component **2520** collectively form a compartment that, when connected, is configured to house an identification element **2420** (e.g., on a flat surface formed on one of the components **2510**, **2520** or collectively; in a compartment formed in one of components **2510**, **2520** or collectively by those components).

(157) FIGS. **27A-D** depict detailed illustrations of the front, bottom, side, and perspective views of an example first component **2510**. The first component **2510** may include a snap mechanism. More specifically, the first component **2510** may include a plurality of snaps **2512** that are configured to be received into corresponding hooks in the second component **2520** (shown below). The first component **2510** may also include a first internal surface **2514**. In some embodiments, the first internal surface **2514** may act as a barrier to help hold the identification element on the surface of a second internal surface **2524** of the second component (shown below), such that it does not peel off. In other embodiments, the first internal surface **2514** may hold an identification element **2420**. In embodiments, the identification element **2420** may be fixed to that surface **2514** using tape, glue, or other mechanism. A bottom surface **2516** opposite the internal surface **2514** may partially define the aperture **2408** into which a connector **200** and its corresponding connection sensor assembly **214** may be inserted, along with side walls **2518**. The first component may have an edge surface **2530** that, along with an edge surface of the second component, forms a groove that grips an interior portion of the connection loop **2402** to hold the housing **2406** in place.

(158) FIGS. **28A-D** depict detailed embodiments of the front, bottom, side, and perspective views of an example second component **2520**. The second component **2520** may include a hook mechanism. The second component **2520** may include a plurality of hooks **2522** for mating with corresponding snaps **2512** of the first component **2510**. The second component **2520** may also

include a second internal surface **2524** to fix an identification element **2420** on top of in conjunction or in the alternative to the second internal surface **2524** of the second component **2520**. In embodiments, the identification element **2420** may be fixed to that surface **2524** using tape, glue, or other mechanism. The second internal surface **2524** may be flat. The second internal surface **2524** may lay on top of the first internal surface **2514** when the first and second components **2510**, **2520** are fixed together, vice versa, or next to one another. An exterior surface **2526** opposite the second internal surface **2524** and side walls **2528** may define the aperture **2408** in collaboration with the corresponding features of the first component **2510**. The second components **2520** may have edge surface **2530'** that, in conjunction with the edge surface **2530** of the first component **2510**, forms the groove that interfaces with the connection loop **2402**.

(159) FIG. **29** depicts detailed embodiments of the snaps on the first component and the hooks on the second component. In that example, the first and second components are fixed together using a cantilever snap-fit joint mechanism. FIGS. **29A-B** depict a detailed embodiment of the snaps **2512**. FIGS. **29C-D** depict a detailed embodiment of the hooks **2522**. The hooks **2522** have a protrusion, which fit into an opening in the snaps **2512**. In some non-limiting embodiments or aspects the first and second components **2510**, **2520** may be made of plastic. The first and second components **2510**, **2520** may be flexible. Further, in some non-limiting embodiments or aspects, the first and second components **2510**, **2520** may be fixed together using a method other than snap-fit, such as an adhesive method, a mechanical method or a welding method.

(160) With reference back to FIGS. **24-26**, the identification element **2420** may be connected to or held within of the housing **2406** held within the connection structure **2400**. In some embodiments, the identification element **2420** may be placed on the internal surface **2524** of the second component, such that the identification element **2420** is enclosed within the housing **2406** when the first and second component are fixed together. The internal surface **2514** of the first component may act as a barrier to help hold the identification element **2420** on the second internal **2524** such that it cannot peel off. In some non-limiting embodiments or aspects, the internal surface **2524** is located proximate to the aperture **2408**. The aperture may be shaped to correspond with the connection area **206**. Furthermore, in some nonlimiting embodiments or aspects, the aperture may be shaped to closely fit the connector **200**.

(161) FIG. **30** depicts an example connection loop before and after a housing is installed therein. In some non-limiting embodiments or aspects, the housing **2406** may be sized to fit within the connection loop **2402**, as shown in FIG. **30**. The edge surfaces **2530**, **2530'** of the first and second components **2510**, **2520** form a outer surface of the housing **2406** that allows for the housing **2406** to rotate within the connection loop **2402**. Therefore, the housing **2406** may rotate with the connector **200**, when the connection structure **2400** is received by the connector **200**. The identification element **2420** may be placed adjacent to the connection sensor assembly **214** on the connector **200**. As the housing **2406** rotates with the connector **200** (e.g., in response to relative movement of the connector **200**), the identification element **2420** follows the connection sensor assembly **214** to maintain a good readability of the identification element **2420**.

(162) As noted above, in embodiments, the housing **2406** may be configured to rotate within the connection loop **2402** based on relative movement of the connector **200**. FIGS. **31A-B** depict the housing **2406** and the connector **200** rotating together from Position I to Position II. Position I may be a starting position when the connector receives the connection structure. When in Position I, the housing **2406** is turned such that the identification element **2406** is disposed between connection point aperture **2408** anchor **2404**. Position II may be any position other than Position I around the loop. In performing actions while connector **200** is connected to the connection structure, the user may move such that the housing **2406** rotates within the connection structure. A closefitting aperture, in accordance with some non-limiting embodiments or aspects, may keep the identification element **2406** close to the connection sensor assembly **214**, facilitating the identification element being more reliably read by a sensor associated with the connector.

(163) Although the disclosure has been described in detail for the purpose of illustration based on what are currently considered to be the most practical and preferred embodiments or aspects, it is to be understood that such detail is solely for that purpose and that the disclosure is not limited to the disclosed embodiments or aspects, but, on the contrary, is intended to cover modifications and equivalent arrangements that are within the spirit and scope of the appended claims. For example, it is to be understood that the present disclosure contemplates that, to the extent possible, one or more features of any embodiment or aspect can be combined with one or more features of any other embodiment or aspect.

Claims

1. A method for detecting connection of a connector to a connection structure, the connection structure having a connection loop that includes a connection point, a connection anchor and a passive identification element disposed between the connection loop and the connection point, wherein the passive identification element is housed in a rotatable component within the connection structure, the connector comprising a frame having an opening and a connection area configured for receiving the connection structure therein and a connection sensor assembly associated with the frame, the connection sensor assembly comprising at least one short-range wireless communication antenna configured for gathering connection data based on detecting a presence or an absence of the passive identification element when the connection structure is received within the connection area of the frame, the method comprising: receiving, with a control device, comprising at least one processor, the connection data gathered by the connection sensor assembly; determining, with the control device, a connection status of the connector based on the connection data; and performing, with the control device, at least one action based on the connection status.
2. The method according to claim 1, wherein the connection point is an aperture formed by the rotatable component and the connection loop.
3. The method according to claim 1, wherein performing the at least one action includes providing at least one of a visual indication, an audio indication, and a tactile indication based on the connection status.
4. The method according to claim 1, wherein detecting the presence of the passive identification element is indicative of a presence of an approved connection structure, and wherein detecting the absence of the passive identification element is indicative of an absence of the approved connection structure.
5. The method according to claim 1, wherein the connection status comprises a connected status indicative of a presence of an approved connection structure, and a disconnected status indicative of an absence of the approved connection structure.
6. The method according to claim 1, wherein the connection sensor assembly further comprises a communication interface, and wherein the at least one action includes transmitting the connection status to a remote device using the communication interface.
7. A fall protection compliance system, comprising: a connection structure comprising a connection loop, a connection anchor that includes a connection point, and a passive identification element disposed between the connection point and the connection anchor, wherein the passive identification element is housed in a rotatable component within the connection structure; a connector having an opening and a connection area configured for receiving the connection structure therein; and a connection sensor assembly associated with the connector, the connection sensor assembly comprising: at least one short-range wireless communication antenna configured for gathering connection data based on detecting a presence or an absence of the passive identification element on the connection structure when the connection structure is received within the connection area; a control device comprising at least one processor programmed or configured

to: receive the connection data from the at least one short-range wireless communication antenna; determine a connection status of the connector based on the connection data; perform at least one action based on the connection status.

8. The system according to claim 7, wherein the connection point is an aperture formed by the rotatable component and the connection anchor.

9. The system according to claim 7, wherein performing the at least one action includes providing at least one of a visual indication, an audio indication, and a tactile indication based on the connection status.

10. The system according to claim 7, wherein detecting the presence of the passive identification element is indicative of a presence of an approved connection structure, and wherein detecting the absence of the passive identification element is indicative of an absence of the approved connection structure.

11. The system according to claim 7, wherein the connection status comprises a connected status indicative of a presence of an approved connection structure, and a disconnected status indicative of an absence of the approved connection structure.

12. The system according to claim 7, wherein the connection sensor assembly further comprises a communication interface, and wherein the at least one action includes transmitting the connection status to a remote device using the communication interface.

13. A fall protection compliance system, comprising: a connection structure comprising a connection loop, a connection anchor, and a rotatable housing, wherein the housing comprises an identification element disposed within the housing; a connector having an opening and a connection area configured for receiving a connection structure therein; and a connection sensor assembly associated with the connector, the connection sensor assembly comprising: at least one short-range wireless communication antenna configured for gathering connection data based on detecting a presence or an absence of the identification element on the connection structure when the connection structure is received within the connection area for the connector.

14. The system according to claim 13, wherein the connection sensor assembly further comprises a control device comprising at least one processor programmed or configured to: receive the connection data from the at least one short-range wireless communication antenna; determine a connection status of the connector based on the connection data; perform at least one action based on the connection status.

15. The system according to claim 13, wherein the housing and the connection loop form an aperture, said aperture fits the connector such that the identification element is adjacent to the connection sensor assembly and wherein the housing fits within the connection loop.

16. The system according to claim 13, wherein the housing further comprises a first component fixed to a second component.

17. The system according to claim 16, wherein the first component and the second component are fixed using at least one of the following fixing methods: snap-fit, adhesive, mechanical and welding.

18. A method for detecting connection of a connector to a connection structure, the connection structure having a connection loop, a connection anchor, and a rotatable housing, the housing comprising an identification element positioned within the housing, a connector comprising a frame having an opening and a connection area configured for receiving the connection structure therein and a connection sensor assembly associated with the frame, the connection sensor assembly comprising at least one short-range wireless communication antenna configured for gathering connection data based on detecting a presence or an absence of the identification element on the connection structure when the connection structure is received within the connection area of the frame, the method comprising: receiving, with a control device, comprising at least one processor, the connection data gathered by the connection sensor assembly; determining, with the control device, a connection status of the connector based on the connection data; and performing,

with the control device, at least one action based on the connection status.

19. The method according to claim 18, wherein performing the at least one action includes providing at least one of a visual indication, an audio indication, and a tactile indication based on the connection status.

20. The method according to claim 18, wherein detecting the presence of the identification element is indicative of a presence of an approved connection structure, and wherein detecting the absence of the identification element is indicative of an absence of the approved connection structure.
